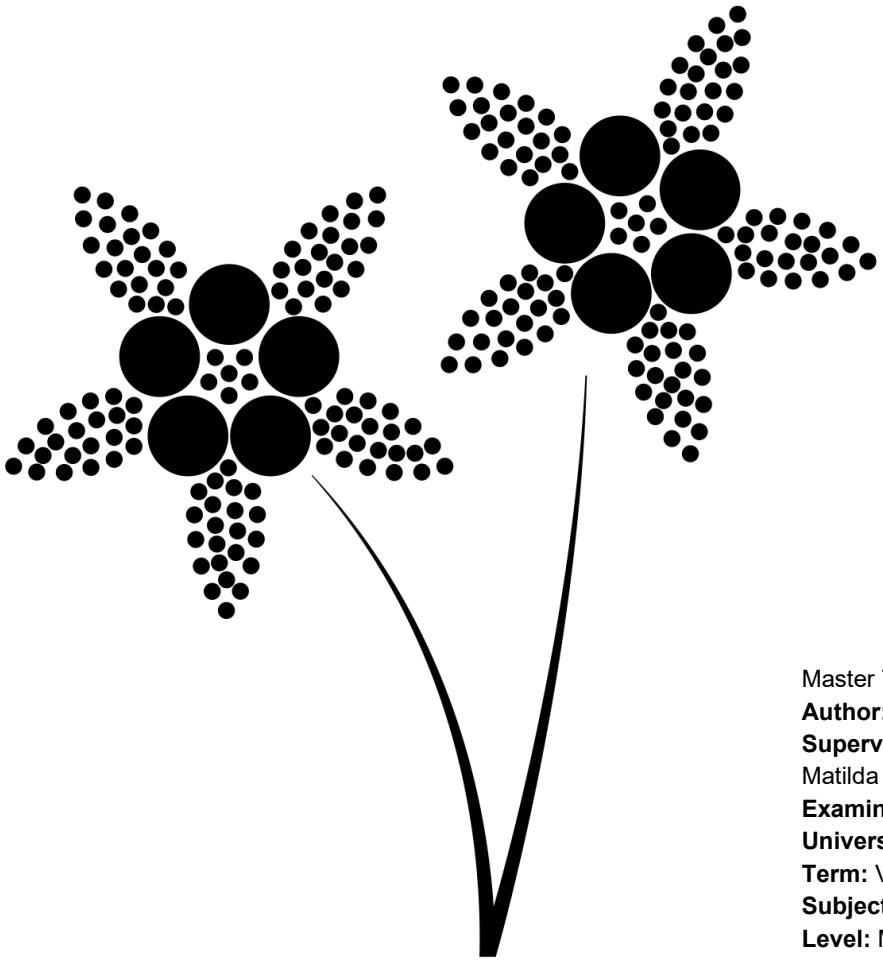
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Agency-Narrativity Mapping Framework

Reading the Gap Between Actual and Perceived Agency in Videogame Episodes



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Abstract

This thesis develops and applies the Agency-Narrativity Mapping framework (ANM), an episode-level analytic framework that tracks how videogames produce narrativity through the mediation of player agency. ANM analyses four semiotic resources: UI feedback, textual framing, spatial layout, temporal design, and renders the gap between actual agency (what the rules permit) and perceived agency (what the player feels they have authored) across four agency dimensions (mechanical, protagonist, operational, authorial). The framework is applied to two bounded episodes from games with strongly contrasting design configurations: the player's first examination of the hanged man in *Disco Elysium* (ZA/UM, 2019) and the tutorial mission "Death of a Showman" in *Hitman: Blood Money* (IO Interactive, 2006). Three participants completed both playthroughs, with screen recordings, Likert-scale instruments, and micro-narrative responses supplying the empirical material. Drawing on multimodal social semiotics, intermedial theory, and game-studies accounts of agency, the analyses find that the agency gap runs in inverted directions in the two cases: *Disco Elysium*'s textual framing inflates perceived authorial agency above the mechanical horizon (Worthen's narrative superposition), while *Hitman: Blood Money*'s UI feedback and traversal-gated spatial layout produce perceived constraint below the mechanical affordance. The comparative discussion demonstrates that ANM holds across games with fundamentally different design philosophies, and that the agency gap operates as a structured network of tensions and alignments. The thesis closes by arguing that agency in a videogame episode is a fluid, multi-layered field surfaced through multimodal-intermedial configurations, of which ANM specifies a tractable portion.

Keywords

videogame narrativity, player agency, multimodal analysis, intermediality, ANM, agency-narrativity mapping framework, *Disco Elysium*, *Hitman: Blood Money*

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1 Introduction

While a portion of a videogame's narrative is conveyed through scripted elements such as dialogue and cutscenes, a significant part emerges during gameplay as the player interacts with the system's rules and connects actions and their outcomes into sequences perceived as causal. This thesis refers to this process as narrativity, defining it, following Elleström, as a meaning-making process in which a story consists of "represented events that are temporally interrelated in a meaningful way" (*Transmedial Narration* 96). This definition situates narrativity within the interpreter who perceives events as temporally and meaningfully connected, rather than within the text or system itself. In the context of videogames, the interpreter is the player, making this definition appropriate for the close analysis of moment-to-moment play undertaken in this thesis and justifying the focus on player perception.

The emergence of narrativity during play depends on legibility: the extent to which the system's feedback and framing enable the player to interpret an occurrence as an event that influences subsequent possibilities. This question runs back to an early disagreement in game studies, the ludology-narratology dispute, over how far videogames should be analysed as abstract rule systems and how far through the concepts of narrative theory. Murray has argued that the dispute was never genuinely joined and is better treated as a heuristic framing than a real opposition, since few scholars ever held that games are simply a kind of story ("The Last Word"). What the dispute named, and what survives it, is a methodological tension any account of game narrativity still must resolve: how a rule-governed system becomes narratively meaningful to the player who operates it. Existing scholarship often addresses only one aspect, either examining narrative content without considering play as an event-generating process or analysing systems and mechanics without demonstrating how they acquire narrative legibility during play.

The analysis is structured by two distinctions, with the second nested within the first. The initial distinction concerns a threshold: among all occurrences during play, only some are designated as significant. Following narratological conventions, the thesis distinguishes between happenings (mere occurrences) and events (occurrences narratively significant within the game's resources). Both the storyworld and player actions generate continuous streams of happenings, but only select happenings cross the threshold to become events. The second distinction addresses the source of these events. Events that have crossed the threshold originate from two sources: represented events, which the storyworld presents as significant (such as the discovery of a body or a character's death), and enacted events, which the player generates through moment-to-moment play (such as crossing a room or selecting a dialogue line).

Agency functions as the analytical bridge between represented and enacted events. Drawing on Norman's distinction between the actual and perceived properties of interaction (9), this thesis differentiates *actual agency*, the system's affordances and

registrations at any given moment, from *perceived agency*, defined as the player's experience of meaningful and consequential action. These two forms of agency do not always align. In some instances, the system may allow actions that the player does not recognise as available, or it may register actions as consequential without the player perceiving their impact. Conversely, semiotic richness can elevate the player's sense of authorship beyond what the system's branching structure supports. Existing scholarship refers to these discrepancies as the *agency gap* (Wardrip-Fruin et al., "Agency Reconsidered"; Thue et al.; Fendt et al.). The relationship between actual and perceived agency, in both aligned and divergent forms, and its impact on narrativity at the episode level, constitutes the analytic foundation of the subsequent chapters.

Examining this relationship in practice necessitates a method that integrates three domains: the design resources that configure actual agency, the semiotic and intermedial cues through which players interpret these configurations as perceived agency, and the events that arise as players construct these interpretations into narratively meaningful sequences. Currently, no existing framework addresses all three domains at the level of episode-based close analysis.

In response, this thesis develops and tests the Agency-Narrativity Mapping framework (ANM), a provisional analytic framework for episode-level analysis. ANM examines how design resources configure actual and perceived agency, how the relationship between these forms of agency shapes narrativity, and how this relationship varies across genres. The analysis centres on two contrasting cases: 1) *Disco Elysium* (ZA/UM, 2019) exemplifies a text-oriented episode in which narrative relevance is established primarily through dialogue, choice prompts, and skill-based commentary; 2) *Hitman: Blood Money* (IO Interactive, 2006) represents a spatial and systemic episode where narrative relevance emerges through movement, timing, detection, and the management of a responsive environment. This contrast tests whether ANM can facilitate comparative episode-level analysis across both genres without erasing their differences.

1.1 Aim and research questions

ANM integrates procedural, multimodal, and intermedial perspectives within a single episode-level analytical instrument, and the framework is operationalised through two contrasting case studies that test whether it holds together as an analytic tool and surface its scope conditions, with applicability beyond the present cases left open for further testing.

Research Questions:

RQ1: How do semiotic resources (UI feedback, textual framing, spatial layout, and temporal design), together with intermedial techniques and procedural techniques, constrain and enable actual agency in each episode, as evidenced in recorded play sequences?

RQ2: How do the semiotic resources, procedural techniques, and intermedial techniques shape perceived agency and the player's construction of narrativity as temporally meaningful event-relations, as evidenced in screen recordings and participant micro-narratives?

RQ3: What recurring alignments and misalignments between actual and perceived agency across the cases suggest directions toward a reusable analytic model for mapping agency–narrativity relations across videogame episodes?

The study addresses these questions through close analysis of screen-recorded play sessions from three participants engaging with two contrasting episodes (*Disco Elysium*, ZA/UM 2019, and *Hitman: Blood Money*, IO Interactive 2006). Analysis applies ANM to map how semiotic resources configure actual agency, with participant micro-narratives (150- to 250-word written reflections each participant produces after playing) and a brief Likert autonomy questionnaire triangulating perceived agency.

1.2 Disposition

Chapter 2 surveys prior research in transmedial and cognitive narratology, multimodal and intermedial media theory, and game studies. It establishes the theoretical framework, introduces the ANM analytical method, and outlines the empirical methodology. Chapters 3 and 4 implement ANM in the analysis of bounded episodes from *Disco Elysium* and *Hitman: Blood Money*, respectively, to address RQ1 and RQ2 through case-study analysis. Chapter 5 offers a comparative discussion of both cases to address RQ3. Chapter 6 concludes by summarizing the findings, evaluating the framework's scope and limitations, and proposing directions for further research.

2 Previous Research, Theory, and Method

This chapter reviews existing scholarship across three fields, transmedial and cognitive narratology, multimodal and intermedial media theory, and game studies, and assembles the conceptual apparatus that ANM operationalises in the analysis chapters. Each field has generated research adjacent to this thesis, but little work has attempted to integrate all three at the level of episode-based close analysis.

2.1 Narrativity across Media: Transmedial and Cognitive Approaches

The ludology-narratology dispute introduced in §1 left a methodological residue that outlasted it. Most scholars now accept that games produce meaning through both systemic and representational resources, yet the tension persists as a practical problem whenever an analyst must decide whether to foreground narrative content or procedural structure. The lack of a unified method for doing both at once is the gap this thesis addresses.

Transmedial narratology offered a way past this impasse by defining narrativity independently of any single medium. The field's central insight is that narrative is a cognitive and communicative process that can be realised through diverse media. Elleström's account of *Transmedial Narration* provides the definition used in this thesis: a story consists of "represented events that are temporally interrelated in a meaningful way" (*Transmedial Narration* 96). Elleström links this interrelation to perceived coherence, arguing that interpreters connect dispersed cues into what he calls "a consistent virtual sphere" even when communication is distributed across media types and moments of access (97). This definition is productive for videogame analysis because it does not require a fixed plot, a singular narrator, or linear sequencing; it only requires that an interpreter perceive events as temporally and meaningfully connected.

Thon's *Transmedial Narratology and Contemporary Media Culture* extends the transmedial-narratological programme to films, comics, and videogames, demonstrating that narratological concepts such as narrator and focalization can be adapted for games provided the adaptation attends to medium-specificity, what Thon calls media-conscious narratology (22). Thon also concedes that text-centred narratology's hypotheses about how recipients understand a storyworld tend to rest on the analyst's own playing experience, and that these questions "must be explored via empirical methods of investigation" that the approach itself does not supply (11). This thesis builds on Thon's media-conscious orientation while taking up that empirical task directly, specifying through participant data the semiotic resources through which narrativity is produced in the particular episodes under analysis.

Jenkins' essay "Game Design as Narrative Architecture" presents an alternative approach to game narrativity through spatial design. Jenkins identifies four modes by which narrative can enter games: evocative spaces that draw on prior narrative associations, enacted narratives structured around the player's movement through contested space, embedded narratives distributed as discoverable information within the environment, and emergent narratives that arise from systems rich with narrative potential (118–130). This typology remains valuable for understanding the macro-organisation of game narrativity, yet it operates primarily at the level of design philosophy and does not address semiotic detail. Jenkins does not consider how specific elements such as UI cues, dialogue options, or timing windows render particular moments narratively legible for individual players. The present thesis extends Jenkins' spatial orientation by treating spatial layout as one of four semiotic resources tracked at the episode level. Nitsche's *Video Game Spaces* provides further analytical detail by distinguishing five layers through which game space operates: rule-based, mediated, fictional, play, and social. His close attention to how games remediate cinematic camera conventions directly informs the intermedial techniques examined in this thesis (15–16; 92–93).

Cognitive narratology provides a complementary foundation. Herman's account of how narrative scaffolds the interpreter's effort to make sense of lived experience supports the decision to collect participant micro-narratives as evidence of narrativization. Each micro-narrative is treated as a structured trace of what the participant regards as an event, a consequence, and a connection (3).

What is missing from these transmedial and cognitive accounts is a method for tracking how narrativity is produced during play, specifically through the semiotic and procedural resources that organise the moment-to-moment relationship between system affordances and player interpretation. Elleström provides the definition, Jenkins and Nitsche offer the spatial architecture, Thon supplies the media-conscious narratological framework, and Herman contributes the cognitive orientation. However, none of these frameworks instructs analysts on how to annotate a bounded segment of recorded gameplay or describe how semiotic resources configure agency and narrativity within that segment. ANM is designed to address this methodological gap.

2.2 Multimodal and Intermedial Perspectives

The preceding section established how narrativity is defined in this thesis, the question that follows is how to describe its production in practice. Multimodal and intermedial research offers the descriptive vocabulary needed to specify how narrativity emerges through the interaction of multiple meaning-making resources.

2.2.1 Semiotic Resources and Multimodal Analysis

The social-semiotic tradition developed by Halliday and extended by Kress and van Leeuwen supplies the basic vocabulary used in this thesis for describing how meaning is produced through multiple resources operating together. Van Leeuwen defines semiotic resources as having "a meaning potential, based on their past uses, and a set of affordances based on their possible uses" that "will be actualized in concrete social contexts where their use is subject to some form of semiotic regime" (*Introducing Social Semiotics* 285). Modes are a particular kind of semiotic resource: Kress and van Leeuwen describe them as "semiotic resources which allow the simultaneous realisation of discourses and types of (inter)action" (*Multimodal Discourse* 21). Each mode has its own affordances that distinguish it from other modes, and modes can be transmedial in that the same mode (writing, image, sound) can be realised through different media. Multimodal analysis studies how meaning is produced through the orchestration of multiple modes simultaneously, attending to how modes interact, reinforce, or create tension with one another, with the meaning of any multimodal ensemble exceeding the sum of its individual modes.

Bateman, Wildfeuer, and Hiippala's *Multimodality: Foundations, Research and Analysis* consolidates existing research by presenting a problem-oriented methodology for applying multimodal analysis across various disciplines and artefact types. The authors' framework analyses communicative situations by identifying canvases and sub-canvases, the modes present within each, and the specific cognitive work required from consumers to engage with the artefact. It distinguishes between linear consumption and ergodic engagement, further categorising the latter as mutable or immutable.

2.2.2 Multimodal Approaches to Videogames

Multimodal approaches to videogames have been relatively rare compared to the volume of game-studies scholarship that treats games through formalist, narratological, or cultural-studies lenses, but the field has coalesced since approximately 2012 through a series of key monographs and articles.

Ensslin's *The Language of Gaming* provided an early introduction to multimodal videogame analysis, and her subsequent *Literary Gaming* proposed a literary-ludic spectrum and a framework of functional ludostylistics comprising four areas: ludology, ludonarratology, ludosemiotics (procedural rhetoric, UI, text and discourse, multimodality), and mediality. Ensslin's literary-ludic spectrum is applicable to positioning *Disco Elysium* (heavy literary-textual emphasis) relative to *Hitman: Blood Money* (heavy ludic emphasis) and is taken up in 5.1.

Hawreliak's *Multimodal Semiotics and Rhetoric in Videogames* offers a comprehensive effort to integrate multimodal analysis with game studies. Drawing on social-semiotic and discourse-analytic traditions, Hawreliak identifies audio, visual, textual, haptic, and procedural modes as key semiotic resources available to game designers and perceived by players. His principal contribution is the argument that procedurality constitutes a distinct

semiotic mode. Procedurality remains underexplored in multimodal studies because it is non-sensory, functioning through rules, parameters, and conditional processes that are not directly perceived by players. It depends on other semiotic modes for reification; a rule becomes interpretable only through the visual, textual, and auditory modes that render it perceivable to the player (124–132). Hawreliak's concept of modal consonance, developed through case studies such as *Mass Effect: Andromeda*, *Verdun*, and *Dys4ia* (156–184), explains how multiple modes can align toward a common experiential goal, enabling the ensemble's meaning-making capacity to surpass that of any single mode. His complementary concept of modal irony (220–246) addresses how modes can work against one another, producing dissonance that opens space for critical or subversive meaning. Both concepts are central to the analysis chapters, clarifying how semiotic resources converge or diverge to produce narratively legible moments. However, Hawreliak's framework is primarily focused on rhetoric and persuasion; narrativity and agency receive comparatively less attention, and his analyses emphasize game-wide rhetorical strategies rather than close readings of specific episodes.

The insight from Hawreliak that is most directly productive for this thesis is that procedurality requires other modes for reification. This offers theoretical justification for the relationship between ANM's three analytic layers: procedural techniques are a distinct layer, but their narrativity effects are always mediated through the semiotic resources that make them perceivable and interpretable. Intermedial techniques similarly operate through these semiotic resources without forming a separate perceptual channel.

While Hawreliak emphasizes rhetorical analysis, Toh's *A Multimodal Approach to Video Games and the Player Experience* provides empirical grounding. Toh, drawing on Kress, defines mode as "a socially shaped and culturally given semiotic resource for meaning-making" (22) and constructs a ludonarrative model based on three relational categories: ludonarrative dissonance, ludonarrative resonance, and ludonarrative (ir)relevance. Each category is further refined into empirically grounded subcategories through participant interviews (chs. 4–6). Toh's empirical methodology, which integrates observation, qualitative interviews, and frame-by-frame analysis of gameplay recordings, closely aligns with the procedure employed in this thesis.

A recent significant contribution is Davidsson's *Animal Encounters in Play*, a multimodal study of the animal-protagonist games *Stray* and *Endling: Extinction Is Forever*. This work draws on theoretical resources also employed in the present analysis, including Elleström's modalities and the social semiotics of Kress and van Leeuwen. Davidsson analyses each game as a set of multimodal configurations and examines how game mechanics both enable and restrict the agency of the controlled character. Two of her observations are particularly relevant to the current framework. First, procedural constraints can conflict with a game's narrative emphasis, as demonstrated by *Stray's* scripted structure, which provides the player with less agency than the narrative suggests (154). Second, *Stray* communicates urgency through its robotic characters without imposing actual time pressure, which Davidsson describes as "a dissonance between the

story's presentation and the player's actual experience" (155). ANM positions such divergences as its central focus, and the temporal example aligns with the distinction between coordination-time and real-world-time that its temporal-design resource is designed to address. The two projects differ in their focal points. Davidsson sets aside player agency to focus on animal agency and the associated interspecies ethical considerations, defining represented agency as a sliding scale that runs from the cipher, an animal-shaped shell without depth, to a creature possessing its own will (163). Her unit of analysis is the whole-game thematic configuration. In contrast, ANM centres on player agency and annotates the actual/perceived relationship at the level of a bounded episode. Considered together, these projects partition the same multimodal landscape between nonhuman and human aspects of agency. Davidsson's identification of presentation-experience dissonance provides independent confirmation, from a distinct type of game, that the agency gap is a pervasive feature of mediated play.

Three additional multimodal studies inform the design of the framework. Bateman, Wildfeuer, and Hiippala dedicate an entire chapter of *Multimodality* to computer and videogames (366–378), analysing turn-based strategy games (*Civilization V*) and real-time first-person games (*Armed Assault 3*) as inherently mutable and ergodic artefacts whose meaning is realised "on a spatiotemporal, interactive canvas" (367). Their analysis conceptualises games as nested communicative situations in which players perform multiple types of ergodic work simultaneously across several sub-canvases: the 3D environment, the interface overlay, and the linear canvas of the sound stream, each carrying different modal configurations that must be coordinated by the player. They draw an explicit distinction between coordinated time (turn-based) and real time, with the two producing different pace and scale of synchronisation across canvases (377). They also identify fade transitions, cutscenes, and depicted media within gameworlds as borrowings whose meanings are inherited from film and other prior media (374). This canvas-decomposition methodology provides the most direct precedent for ANM's episode-level resource annotation. Wildfeuer and Stamenković extend this approach to game tutorials using dynamic discourse semantics (28–51), while Stamenković and Wildfeuer's semiotic inventory of *GTA V* (259–280) annotates semiotic elements across successive gameplay stages, offering the most concrete example of the moment-by-moment annotation proposed by ANM. Pérez-Latorre, Oliva, and Besalú introduce a four-level analysis model that combines social semiotics and procedural rhetoric, applied to *The Last of Us*, whose multi-layered architecture aligns productively with ANM's analytic layering.

Disco Elysium has received increasing scholarly attention. Bódi and Thon's four-dimensional typology of player agency, as applied to the game (157–190), represents the most directly relevant precedent for ANM's analysis and is discussed further in section 2.3.6. A 2021 special issue of Baltic Screen Media Review, dedicated entirely to *Disco Elysium*, examines the game's polyphonic subjectivity through Bakhtin (Vella and Cielecka 91–102), as well as its Gothic, posthuman, and existentialist aspects. Makai

applies Elleström's modalities model to the game's intermedial world-building. No comparable semiotic, multimodal, or intermedial analysis of *Hitman: Blood Money* has been identified in the literature; this represents a gap that ANM aims to address.

2.2.3 Intermediality and Media-Sensitive Narratology

Intermediality, as defined by Bruhn and Schirrmacher in the introduction to *Intermedial Studies*, refers to "the interaction of similarities and differences between media and the changes that may occur in communicative material when it is transported from one media type to another" (3). Intermedial studies examine the relationships between media products and media types, focusing on how these relationships themselves generate meaning. In this thesis, intermediality serves as the analytic lens through which to specify how conventions adopted from other media function as design techniques within videogames.

Two approaches within intermedial research are particularly pertinent to this thesis, each advancing the field in distinct directions. Rajewsky offers a restrictive, literature-oriented conception in which intermediality denotes specific relationships between distinct media products, categorised into three subtypes: medial transposition, media combination, and intermedial references ("Intermediality, Intertextuality, and Remediation" 51–53). In contrast, Elleström presents a broader, communication-theoretic perspective, positing that intermediality emerges from the inherently multimodal nature of all media. He describes media products in terms of four modalities (material, spatiotemporal, sensorial, semiotic) and characterises intermediality as "media intermultimodality," a complex set of relations among media that are more or less multimodal in various ways (*Beyond Media Borders* 85). This thesis combines the two. Rajewsky's intermedial reference supplies the type of relation tracked in the analyses: cinematic staging, theatrical blocking, montage-like pacing, and musicological scoring as they are incorporated into game design. Elleström's modalities offer the descriptive tools for articulating how these references function at the level of modal configuration within specific game artefacts.

Rajewsky's three subcategories include medial transposition (the transformation of one media product into another), media combination (the presence of multiple media in their own materiality within a single product), and *intermedial references* (a media product employing its own means to evoke or imitate elements of another medium) ("Intermediality, Intertextuality, and Remediation" 51–53). The third subcategory is most relevant to this thesis. Rajewsky, citing Heller, characterises the author of an intermedial reference as one who "writes as if he had the instruments of [the other medium] at his disposal, which in reality he does not" (54–55). This "as if" formulation was originally developed for literary contexts, where a novel might describe filmic operations using only verbal means. Computer games, however, complicate this distinction: they can edit moving images, cut between shots, employ camera movement, dissolve, and montage in real time through computational processes. Rajewsky's framework remains illuminating for games by highlighting the inheritance of convention, even when the operations themselves are native to the game medium. Techniques such as a cinematic dolly-in, a fade-to-black, a

leitmotif in a game's score, or theatrical blocking of bodies on screen derive their meanings from the prior media (film, music, theatre) that established these conventions. The intermedial reference thus operates at the level of convention, even when the operation is native to the referencing medium.

Elleström's modalities model offers a different point of entry into intermedial analysis. He shifts the focus from relations between distinct products to the modal constitution of media products themselves, distinguishing four modalities that condition meaning-making in any media product: the material modality (the physical interface through which signs are transmitted), the spatiotemporal modality (how the media product is organised in space and time), the sensorial modality (which senses are engaged and how), and the semiotic modality (what kinds of sign relations are operative) (*Beyond Media Borders* 46–53). On this view, intermediality is a constitutive feature of media as such: every media product is itself a configuration of modalities, and intermediality is the study of how those configurations relate to one another (85). Hammeleff Jørgensen has directly operationalised Elleström's framework for game analysis, proposing a revised four-modality system (structural, material, sensorial, semiotic) and demonstrating its application to *FRAMED*, a game that exploits intermedial affinities between comics and games ("Media and Games" 1–10). This revised modality system is taken up in 3.4 and 4.3 to specify what intermedial reconfigurations adjust at scripted moments. Makai's practical intermedial toolkit, building on Elleström and applied to *The Witcher 3*, demonstrates the practical operability of intermedial analysis for games (Bruhn and Schirrmacher 309–328).

A specifically game-relevant extension of intermedial analysis comes from Keylin, who proposes the concept of a "mediality of own action" to name the constitutive role of user activity in digital media (1455–1459). In media where the user's input generates the audiovisual flow (interactive sound, networked communication, videogames), what the perceiver encounters is shaped by their own actions; the perceiver is also a kind of producer. For videogames, this concept names the situation ANM is designed to analyse: the player's input is one of the conditions of the perceived configuration, and what a player perceives as the game's semiotic surface is partially the product of their own play. This generative dimension distinguishes videogames from the audiovisual media (film, television) whose intermedial conventions they nevertheless inherit.

The intermedial techniques tracked in ANM are specified by their media of origin, in line with Rajewsky's caution against media blindness ("Potential Potentials" 24–25): cinematic staging refers to conventions borrowed from film, including camera framing, shot composition, depth-of-field attention cues, and cut-scene editing rhythms; theatrical blocking refers to spatial composition conventions associated with stage drama, including proximity, facing, elevation, and sightlines as indicators of relational significance; montage-like pacing refers to editing and juxtaposition conventions drawn from film and comics, where the sequencing and rhythm of discrete moments creates anticipation, contrast, or causal implication; and musicological scoring refers to conventions drawn

from film scoring and concert music traditions, including leitmotif, dynamic soundtrack layering, and the deployment of silence.

The edited volume *Intermedia Games: Games Inter Media* (Fuchs and Thoss) is the most important collection on games and intermediality, with contributions on cinematic remediation, montage, camera conventions, and comics aesthetics in games. Together with King and Krzywinska's earlier ScreenPlay: Cinema/Videogames/Interfaces, Galloway's analysis of the first-person shot's migration from cinema to games (*Gaming* 39–69), and Klevjer's work on cutscenes as intermedial phenomena that create a "parallel cinematic space" (301–309), this body of scholarship confirms that games are densely intermedial and that their intermediality is functionally significant for how players experience narrative, space, and agency. Klevjer's parallel cinematic space concept anchors the analysis of DE's vomiting close-up scene and HBM's split-screen.

Summers's *Understanding Video Game Music* provides the most sustained analysis of musical intermediality in games, examining leitmotif, dynamic music systems, and the diegetic/non-diegetic distinction. His account of alert-state scoring in stealth games, where the music shifts with the game's detection state, is the precedent the present analysis draws on (122–123).

An important bridge between multimodal description and narrative analysis comes from Punday's *Playing at Narratology*. In his chapter "UI Time and the Digital Event," Punday argues that interface structures can themselves function as narratological arguments: when an interface waits indefinitely for input, escalates alerts over time, or punctuates action with confirmation cues, it implicitly models what counts as an event and when an outcome counts as a consequence (75–92). Punday's UI time concept supports the decision in this thesis to treat UI feedback and temporal design as conditions of narrativity, without reducing them to neutral delivery mechanisms, and is operationalised at 3.3.1 and 4.2.1.

2.3 Agency, Rules, and Feedback: Game-Studies Accounts

Describing how games produce narrativity through play also requires concepts that can account for rules, representations, and feedback loops, the very mechanisms through which a game structures action and makes it interpretable. Game studies have generated substantial scholarship on these questions, though the field has mostly theorised agency and meaningful play as properties of design and has paid less attention to how multimodal configurations shape them.

2.3.1 Meaningful Play and Legibility

Salen and Zimmerman's account of meaningful play establishes a baseline condition: for play to be meaningful, the relationship between player action and system outcome must be both discernible (the player can perceive what happened) and integrated (the action matters for the larger experience) (34). This formulation motivates the thesis's focus on legibility as a semiotic condition. However, Salen and Zimmerman do not specify how discernibility and integration are achieved (through which semiotic resources, intermedial conventions, or procedural structures a game makes action–outcome relations perceptible). ANM operationalises their condition by tracking how each of the four semiotic resources contributes to making the action–outcome relation discernible and integrated in specific ways.

Murray's widely cited definition of agency as "the satisfying power to take meaningful action and see the results of our decisions and choices" (*Hamlet on the Holodeck* 126) has shaped the field considerably and has also drawn criticism for conflating what a system can do with what a player feels. This thesis responds by drawing a line between actual agency and perceived agency. Norman's separation of actual and perceived properties of interaction in design theory supports this distinction (9).

2.3.2 The Actual/Perceived Agency Gap: Theoretical and Empirical Research

Mateas and Stern's distinction between local and global agency offers a more nuanced framework for analysing player interaction. Local agency is defined as the system's immediate, context-specific, and meaningful responses to player actions, while global agency refers to the extent to which player actions shape the overall trajectory of the experience (4–5). Their research on the interactive drama *Façade* demonstrates that vivid local feedback can maintain a strong sense of agency, even within a tightly controlled global structure. Wardrip-Fruin, Mateas, Dow, and Sali extend this from a different angle, reconceptualising agency as a phenomenon involving both player and game, occurring when the actions players desire are among those they can take as supported by the underlying computational model, rather than being solely an audience experience or a structural property of the work ("Agency Reconsidered"). According to this view, players must shift from their initial expectations to an understanding of the system's model, and significant violations of these expectations reduce the sense of agency. Dow's empirical study of *Façade* further clarifies that a more immersive and realistic interface heightened player expectations and diminished perceived agency, even when the system's affordances remained constant. This finding demonstrates that the possibilities a system offers and those a player perceives can diverge, and this divergence defines the concept of the agency gap.

A growing body of empirical work lends weight to these theoretical arguments. Fendt et al. built a non-branching game designed to simulate branching choices and found that players reported comparable feelings of agency in branching and non-branching versions,

suggesting that semiotic cues like choice presentation and feedback framing can generate the experience of agency even when actual branching is absent (114–125). Thue et al. draw a substantively similar distinction, separating the agency a game gives the player, the objective ability to alter the experience, from the agency each player perceives, the subjective sense of that ability (91); later literature labels these "theoretical" and "perceived" agency (Day and Zhu). This formulation is the clearest terminological precedent for ANM's own distinction. Carstensdóttir, Kleinman, Williams, and Seif El-Nasr arrived at complementary findings through an interview study: their participants' sense of agency depended on social investment, genre conventions, and prior playthroughs they reasoned against, evidence that perceived agency is constructed through interpretive and semiotic processes.

2.3.3 Operational Logics, Procedural Representation, and System-Level Analysis

Mateas and Wardrip-Fruin's concept of operational logics provides a way to bridge system affordances and meaning-making at an abstraction level between individual rules and overall design philosophy. Operational logics are computational strategies of representation that simultaneously address how games operate procedurally and how they communicate these operations to players (2–8). In ANM, operational logics are identified as part of the procedural techniques layer: the skill-check system in *Disco Elysium* and the disguise/suspicion system in *Hitman: Blood Money* are operational logics that determine what counts as an action and a consequence within their respective episodes.

Bogost's concept of procedural rhetoric, defined as "the art of persuasion through rule-based representations and interactions" (*Persuasive Games* ix), establishes that game rules constitute a mode of expression distinct from textual, visual, and auditory modes. Hawreliak extends this to the multimodal framework by arguing that procedurality is a semiotic mode that satisfies the criteria for mode status. Bogost's concept of the procedural enthymeme, where the player "literally fills in the missing portion of the syllogism by interacting with the application" (34), parallels how players in ANM's framework interpret system affordances through the semiotic cues available to them.

Sicart's *Beyond Choices* argues for the importance of aggregation over branching in ethical gameplay, proposing that accumulated small decisions better simulate ethical complexity than isolated binary choices (105). This distinction informs how actual agency maps onto perceived agency and is taken up in 4.1 in the analysis of the Silent Assassin rating system.

2.3.4 Cybertextuality, Ergodic Participation, and Temporal Conditions

Aarseth's *Cybertext* perspective shifts attention from narrative as a finished message to textuality as a communicative machine comprising a material medium, a system of verbal signs, and a human operator, within whose triad the text "takes place" (21). Cybertexts are characterised by feedback loops in which the operator's actions alter the significance and perceived teleology of what is encountered (18–19). Aarseth's distinction between "what was being read" and "what was being read from" (3) is central to this thesis: in a cybertext, the reader is constantly reminded of inaccessible strategies and paths not taken, and each decision makes some parts of the text more, and others less, accessible (3–4). The texton/scripton distinction (62) names this structural condition and is operationalised throughout 3.3 and 4.2 to describe how each player's build or trajectory converts a different subset of the system's content into accessible narrative material. Aarseth's analysis of aporia and epiphany as fundamental dynamics of ergodic art ("Aporia and Epiphany" 31–41) describes moments where actual and perceived agency maximally diverge or align; ANM tracks these dynamics through its semiotic resource annotations. The concept of ergodic literature is foundational but does not itself provide a method for analysing how specific semiotic resources configure the ergodic process. His typology describes functional differences between textual machines at the structural level; ANM addresses the missing layer by specifying the semiotic resources through which ergodic participation becomes narratively legible.

Juul's account of videogames as constituted by both real rules and fictional worlds maintains the dual character of play. His distinction between play time and fictional time clarifies a basic temporal condition of agency: fictional time is not fully predetermined at the moment of play, and player actions have a dual status as both real input and fictional event (199–220). This distinction becomes central in 3.3.4 and 4.2.4, where pacing, delay, and time pressure are treated as narrative conditions that shape which causal connections players can plausibly make.

Galloway's *Gaming: Essays on Algorithmic Culture* contributes a four-quadrant typology of gamic action that cuts across these concerns: diegetic operator, diegetic machine, nondiegetic operator, and nondiegetic machine (1–38). The typology makes explicit that different gameplay elements operate under fundamentally different agency conditions: the ambience act in which the idling game runs itself (a diegetic machine act) requires no operator at all, while active play (diegetic operator acts) foregrounds operator input, and head-over display (HUD) elements (nondiegetic machine acts) mediate agency in ways quite unlike pause-menu choices (nondiegetic operator acts). Used alongside ANM's semiotic resource analysis, this typology helps specify the agency conditions under which particular intermedial techniques are at work.

2.3.5 Ludonarrative Dissonance as Semiotic Breakdown

Hocking's concept of ludonarrative dissonance, the conflict between the narrative told non-interactively and the narrative told through gameplay, has been widely adopted in game criticism. This thesis operationalises the concept through Toh's empirically-grounded subcategories of ludonarrative dissonance, resonance, and (ir)relevance (2.2.2) and through Hawreliak's modal irony (220–246), which provides a more precise analytical vocabulary distinguishing productive dissonance, where tension between modes generates critical or subversive meaning, from unproductive incoherence, where modes simply fail to align.

2.3.6 Agency in Case-Study Games

Disco Elysium has attracted specific scholarly attention to agency. Bódi and Thon's four-dimensional agency typology (spatial-explorative, temporal-ergodic, configurative-constructive, and narrative-dramatic), applied to *Disco Elysium*, is the most directly relevant precedent for ANM's analysis (157–190). Their finding that the game produces narrative-dramatic agency primarily through textual and interpretive means supports the thesis's expectation that textual framing and UI feedback will be the dominant semiotic resources in the *Disco Elysium* episode.

Bódi expands this typology into a comprehensive monograph, *Videogames and Agency*. The second chapter defines agency as a possibility space determined by a game's affordances and limitations (42). This approach excludes player interpretation and treats agency as a property of game design, realized across multiple playthroughs (42). The book's conclusion reiterates this definition as "the possibility space for avatar action, as afforded and limited by game design" (207). While Murray's influential definition links agency to narrative satisfaction, Bódi critiques this perspective for reducing the gameplay experience to narrative (16). Instead, Bódi proposes a design-oriented heuristic: "on what terms does the game's design allow or constrain the player's freedom to act via the avatar?" (43). For ANM, conceptualizing agency as a design affordance aligns with the thesis's emphasis on actual agency as a system-level property. By excluding player interpretation, this approach specifically anchors the actual side of the actual/perceived distinction.

The game's polyphonic narrative structure has been described as a synthetic genre in which interactive literature, computer role-playing games (CRPGs), and virtual book-games interact, with the player's internal dialogue with the protagonist contributing to the polyphony alongside the game's own voices (Morozova 86–97); this account informs the analysis of skill voices in 3.4. Pape's analyses of stealth games (*Aesthetics of Stealth*; "Moving in Stealth") are similarly taken up in Chapter 4 to specify HBM's procedural and intermedial techniques. No comparable agency-focused analysis of *Hitman: Blood Money* was found in the scholarly literature, confirming that the game represents an underexplored case for the kind of analysis ANM proposes.

2.4 The Agency-Narrativity Mapping Framework

The three bodies of work discussed above each help explain how games create stories through player agency. However, none of them show how a specific game creates this effect during a particular segment of play. Research on the agency gap has shown that there is a real difference between what a system allows and what a player feels they have done, and this gap has been measured (Wardrip-Fruin et al.; Thue et al.; Fendt et al.; Carstensdóttir et al.). But these studies have not linked this gap to the design choices that influence it. As a result, while we know the gap exists and can estimate its size, we still cannot say which features of a game cause it, or why two games might create opposite effects. Agency is still studied either as an abstract system property or as an overall impression of a game, without a way to analyse it at the level where design decisions matter. This gap shows up in three specific ways.

First, and most consequentially, the actual/perceived agency gap has not been linked to the specific semiotic resources responsible for its production. Establishing this connection would transform the gap from a phenomenon observable only after gameplay into one that can be analysed as it emerges, thereby making the concept applicable to close reading, game design, and user research. Second, no study has annotated the semiotic resources of a text-rich game such as Disco Elysium at the level of a single, bounded episode, which is the scale at which this mechanism becomes discernible. Third, no existing research has applied a close multimodal-intermedial analysis to any Hitman game, leaving the agency design of this prominent stealth franchise, constructed upon the very resources such a method would examine, unexplored at this level.

ANM is built to close these three gaps together, and in doing so to make the agency gap analysable as something a game produces through specifiable resources, at a grain fine enough to compare how different designs produce it. The framework translates the conceptual apparatus developed in sections 2.1 to 2.3 into a three-layer analytic architecture for episode-level close analysis. The first layer tracks four semiotic resources: UI feedback (continuous and event-bound interface states, including the multimodal channels that render state changes legible to the player), textual framing (dialogue, captions, descriptive prose, in-system text), spatial layout (the rendered environment as a field of action, including conditional embedding and camera positions that structure spatial reasoning), and temporal design (the temporal logic of the episode, including coordination-time, real-world-time, and the structuring of timing windows). These four resources were selected because they are present and trackable across both case studies, sufficiently specific to support moment-by-moment annotation, and capable of capturing how different modal configurations generate narratively distinct effects (2.2.1). The second layer tracks intermedial techniques in Rajewsky's restrictive sense of intermedial reference (2.2.3). The third layer tracks procedural techniques: operational logics as defined by Mateas and Wardrip-Fruin (2.3.3), the cybertextual feedback structure described by Aarseth (2.3.4), and the rule/fiction duality developed by Juul

(2.3.4). Collectively, these three layers specify what renders actions perceptible, interpretable, and connectable as events for a particular player at a particular moment.

The distinction between actual and perceived (2.3.2) serves as the central mechanism for applying the framework to bounded episodes. The relationship between these two elements, whether aligned or divergent, enables ANM to track the production or disruption of narrativity during play. Bódi's possibility-space definition provides the foundational system-side anchor.

When applied to a bounded episode, the framework functions across three layers. Layer 1 (semiotic resources) determines what the player can perceive and interpret. Layer 2 (intermedial techniques) incorporates conventions from other media to guide attention and consequence. Layer 3 (procedural techniques) establishes what constitutes an action and a consequence within the episode. The relationship between actual and perceived agency emerges from the interaction of these layers, including the specific forms the agency gap assumes and the ways in which these forms influence the production of narrativity.

2.5 Empirical Method

Three adult participants (18+) were recruited through channels in which the researcher had no personal or dependency relation. Recruitment was situated in Poland. Participants used their own PCs and usual input setup (keyboard, mouse, one monitor, headphones or monitor speakers). Differences in hardware and settings are noted in the analysis only where they plausibly affect UI feedback, timing, or the readability of interface cues.

The analysis of each episode in Chapters 3 and 4 proceeds in two passes. First, the recording is annotated for salient moments where agency and narrativity hinge on a cue, constraint, or shift in system state. A moment is defined as a short segment in which an intention forms, an action is taken, and feedback stabilises into an interpretable consequence. Second, each moment is mapped onto the four semiotic resources and discussed in terms of actual agency (what actions were afforded and constrained) and perceived agency (what the recording and participant account suggest seemed possible or consequential). Each episode yields an analytic memo and a summary matrix recording moment sequence, dominant semiotic resources, actual-agency affordances and constraints, perceived-agency cues from the micro-narrative and questionnaire, and the inferred narrative connection. Cross-case synthesis in Chapter 5 compares matrices to find recurring patterns and divergences. The synthesis supports directions toward a more stable model without making claims of generalisability.

Each participant played two episodes remotely on their own PCs: the opening sequence through the first crime-scene investigation in *Disco Elysium*, and "Death of a Showman" in *Hitman: Blood Money*. Each play session was screen-shared through a private Discord call (used in preference to Zoom because of latency issues with the latter's screen-

sharing). The call transmitted the gameplay feed only; microphones remained muted and there was no communication between researcher and participant during play, reducing the risk that researcher cues would shape attention or decision-making. The researcher recorded the shared gameplay window using OBS Studio, capturing visuals, visible UI, and in-game audio. Restarts after failure were allowed; repeated reload cycles intended to optimise outcomes were avoided so that consequences remained legible within the episode. After each episode, participants completed two short tasks: a brief autonomy questionnaire using Likert-type items on a 1–7 scale (strongly disagree to strongly agree) targeting felt choice, control, and consequence (the full item set and scores are reproduced as Appendix C); and a 150- to 250-word micro-narrative describing the episode as they experienced it, focusing on what felt consequential, what felt constrained, and how actions and outcomes connected into a narrative sequence (prompt templates are reproduced as Appendix D). These participant texts function as triangulation material for perceived agency and narrativity alongside the researcher's close analysis of the recordings, treated as records of how players construct event-relations from their gameplay. This design follows the triangulation logic Toh applies to player-experience research, combining gameplay recordings, observation, and the players' own accounts so that the analyst's reading of a recording is checked against the player's reported experience rather than standing alone (5).

Personal data is minimised and treated confidentially. Published material in the thesis is anonymised. Consent documentation and the pseudonymisation key are stored separately from research materials. Given the limited number of participants, findings are presented as analytically grounded understandings of how these specific episodes mediate agency and narrativity; statistical generalisability is not claimed.

3 *Disco Elysium* Analysis

3.1 Introduction

Disco Elysium (ZA/UM, 2019) is a detective role-playing game in which an amnesiac officer of the Revachol Citizens Militia investigates a hanging in the working-class district of Martinaise. The game's narrative is discovery-driven: the player learns the protagonist's identity, the circumstances of the case, and the wider social and political world of the city through dialogue choices, environmental interactions, and the commentary of the protagonist's twenty-four internalised skill voices, each speaking with its own distinct register. This chapter analyses how the four semiotic resources tracked by ANM configure actual and perceived agency during the player's first crime-scene investigation in the game: the examination of the hanged man in the yard behind the Whirling-in-Rags hostel. The analysis takes Bódi and Thon's design-level classification (2.3.6) into episode-level semiotic annotation within a bounded segment of recorded play and finds that the agency gap takes three distinct forms in the episode, varying across the three participants.

3.2 Episode Overview and Procedural Techniques

The hanged man episode is structured by four operational logics in Mateas and Wardrip-Fruin's sense (2–8): the skill-check system, the dialogue branching structure, the internal monologue system (including the thought cabinet), and the dual health/morale resource system. An operational logic, in their formulation, is a computational strategy of representation that simultaneously addresses how a game operates procedurally and how it communicates those operations to players. Bogost's procedural enthymeme names the inference structure (*Persuasive Games* 34): the player encounters a rule's visible effect, fills in the underlying rule by interpretive completion, and acts on the inferred rule until contrary evidence forces revision. Together, the four logics constitute the procedural baseline against which 3.3 reads what the semiotic resources surface or conceal.

Skill checks resolve through a simulated dice roll combining the player's stat allocation with a randomised element, with two types: white checks (retriable once new information or items are acquired, or skill points invested) and red checks (single attempt, no retry). A hover popup shows the relevant skill, current level, colour-coded verbal assessment, numerical percentage, white/red designation, and dice diagrams (fig. 3.3). Dialogue branching presents context-dependent option sets in which some lines are freely available, some are marked skill checks, and some are entirely invisible to characters whose relevant skill falls below threshold. Twenty-four skills, each personified as an internal voice with a distinct register, narrate process and outcome through active checks (player-invoked) and passive interjections (system-invoked based on stat level and context). Two resource bars sit in the bottom-left corner: Health (physical) and Morale (psychological), each depletable to zero ending the game and restorable by consumable items (fig. 3.4). Health depletes through physical events such as failed Endurance checks and environmental damage; Morale depletes through psychologically costly events such as humiliating dialogue exchanges and emotionally heavy choices. The playthrough ends when either bar reaches zero, and the game uses "morale death" for the psychological case and "health death" for the physical one. The thought cabinet operates on a longer timescale than this episode engages and contributes without being foregrounded.

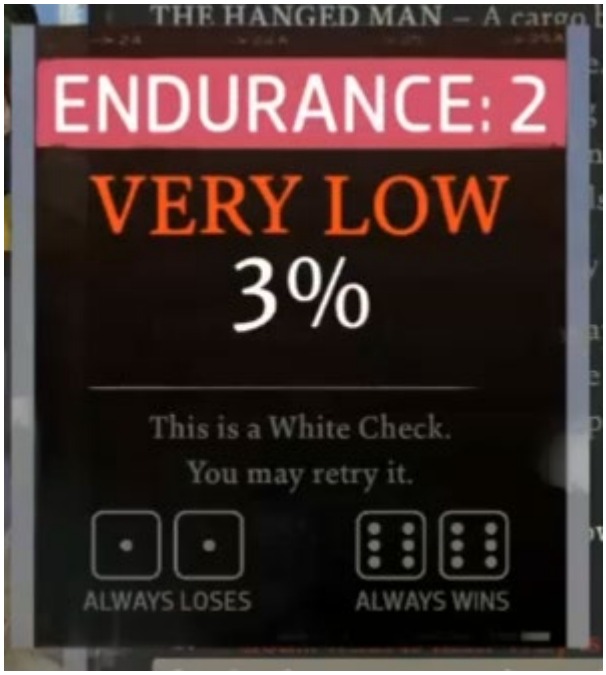


Fig. 3.3 *Disco Elysium*'s white skill-check window.

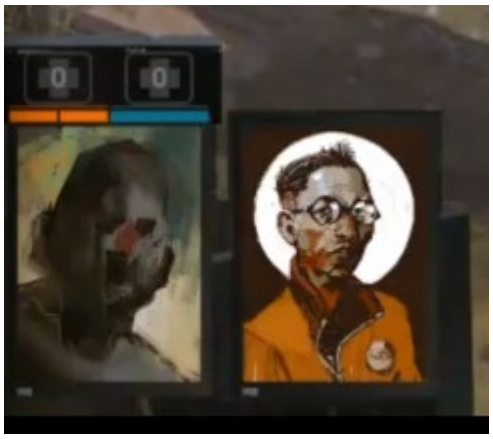


Fig. 3.4 *Disco Elysium*'s top left health (orange) and morale (blue) bars, and underneath the protagonist's and Kim Kitsuragi's portraits.

Player interaction with these systems diverged sharply across the three participants (see Appendix A for a detailed matrix). All three reached the crime scene with character builds that precluded success on the central Endurance white check at legendary difficulty, reflecting ZA/UM's stated philosophy that mistakes reshape future play (Kurvitz; Keenan; Stone). What differed was what the four systems each made perceptible to each player. P01 succeeded on a Visual Calculus check at the hanged man's feet and had that skill dominate the subsequent investigation; P02 never encountered that interaction at all, since their Perception held the footstep marks and broken fence below the threshold at which the system rendered them as interactable. P02 saw a Physical Instrument red check during the Cuno conversation that the other two never saw as

available. P02 also died twice from Health depletion before reaching the crime scene through the silent-warning condition; P03 died from Morale depletion during the Cuno conversation, with the system treating dialogue choices as psychologically costly enough to kill the protagonist.

The underlying event sequence was the same for all three participants; what differed across them was the semiotic configuration through which each experienced it. Bódi and Thon's configurative-constructive dimension covers the agency a game affords through configuring the avatar and (re)constructing its spaces (160). They divide it in two: configurative agency, the shaping of the avatar, and constructive agency, the altering of the game's spaces (162–163). *Disco Elysium* largely constrains the constructive half, since the player cannot alter the yard or the world, but affords a fairly high degree of the configurative half: the build, the distribution of points across the protagonist's skills, is set at character creation and developed through play (167). Character creation therefore happens before the narrative begins and determines both what the player can do and what appears as possible to them. It is through this conditional visibility that the configurative-constructive dimension feeds into perceived agency: the build decides which skill voices can speak and which interactables the system will register, so an act of configuration carried out before play conditions what the later episode can make perceptible.

3.3 Semiotic Resource Analysis

The analysis distinguishes several dimensions of agency that operate simultaneously in the episode. Mechanical agency names what the rules permit at any given moment. The concept draws on Bódi, whose second chapter frames agency as the possibility space for avatar action that a game's design affords and limits (Bódi, Videogames and Agency 42). Protagonist agency names what Harry can do as a character within the diegetic fiction. Player agency operates in two registers across these: operational agency, the act of selecting and executing moves the system permits; and authorial agency, the player's sense of directing the narrative through those moves. The four semiotic resources address each of these dimensions to different degrees.

3.3.1 UI Feedback

Three elements organise UI feedback in this episode: the text panel's event-bound bracketing of interaction, the nondiegetic-machine-act status of the UI's most salient communicative events, and the modal consonance produced when those events resolve. Their multimodal orchestration concentrates the system's communicative force at procedural moments: nondiegetic visual events synchronise with diegetic visual, auditory, and textual signals to produce experiential density at the resolution of checks, while the continuously visible Health and Morale bars remain in a discernible-but-unintegrated condition. The configuration makes the player's operational agency legible at the moment of resolution (the dice roll lands, the body responds), while leaving the mechanical agency horizon (the resources that will permit or foreclose future moves) communicated

as state without urgency, the structural condition for the unread-warning form of the agency gap.

The interface comprises four spatially distinct zones (fig. 3.4, 3.6). At the bottom left of the screen, two portrait frames display the protagonist and Kim Kitsuragi (the protagonist's work partner), portraying protagonist's health bar (physical) and a morale bar (psychological) above the portraits; these bars are the episode's only continuously visible state indicators. At the bottom right, four icons provide access to the character sheet, inventory, journal, and thought cabinet, alongside a clock display, day counter, currency, and equipped-item slots. At screen centre, a non-transparent text panel (fig. 3.5) opens on the right when the player initiates an interaction with a non-player character (NPC), an object, or an environmental detail, delivering dialogue, internal monologue, environmental description, and player choices through the same surface. The skill-check popup, the success/failure colour-wash, and the dice display appear only at procedural moments and dissolve once the resolution is acknowledged.

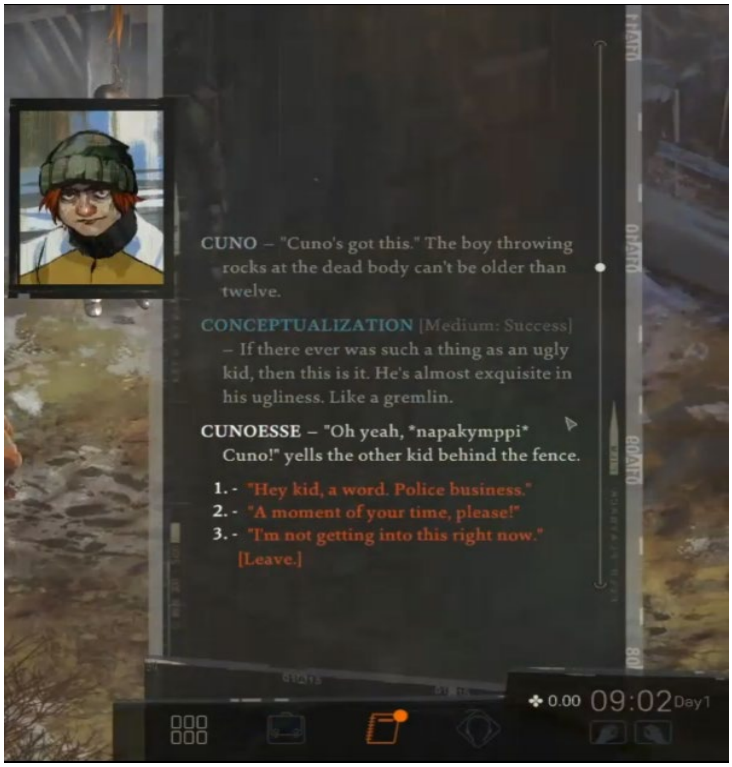


Fig. 3.5 *Disco Elysium*'s non-transparent interaction window.



Fig. 3.6 *Disco Elysium*'s four interactable UI interfaces accompanied by currency, time of day and two hand equipable slot visualizations in the right corner.

The text panel illustrates what Punday calls UI time: a third temporal layer in digital narrative, separate from story time (the temporality of the events represented) and discourse time (the time the work takes to present them), and defined by what the program is waiting for. A computing device sits idle, watching for an event it has been built to recognise, and responds only when the user triggers one (75–92). The DE panel runs on this logic. It is not on screen by default; the player opens it by interacting with an NPC, object, or environmental detail, and it closes once that exchange concludes. Each interaction is thus a self-contained event, and between interactions the panel registers nothing. The continuously visible skill icons and resource bars belong to a different register of UI feedback. They stay on screen throughout, reporting the protagonist's condition whether or not an interaction is underway. Their job is to show state, so they do not open and close around events the way the panel does.

The skill-check popup, colour-wash, and check banner are nondiegetic machine acts in Galloway's sense: they are system-side reports addressed to the player. The protagonist receives no equivalent of address. Only the player sees the red wash that follows P01's failed Endurance check; the game does not register the protagonists, Harry's, perception of any of the UI elements players read. This positions the player as the operator of a computational system being briefed on its current state, the form of communication Wardrip-Fruin and colleagues theorise as expectation management ("Agency Reconsidered"). The popup's single frame consolidates the relevant skill, percentage, verbal assessment, check type, and dice thresholds into the information needed to evaluate an action before committing to it. The same nondiegetic logic governs the post-resolution reporting: dice values, "CHECK FAILURE" / "CHECK SUCCESS," colour-wash (fig. 3.7, 3.8). The interface positions the player as a planner with system-level information the protagonist does not have.



Fig. 3.7 *Disco Elysium*'s skill-check failure display.



Fig. 3.8. *Disco Elysium*'s skill-check success display.

When P01's Endurance check fails at the hanged man, four channels arrive together. The colour-wash and check banner are nondiegetic visuals. The forced vomiting animation, with altered camera framing and visceral sound design, is diegetic visual and auditory. Endurance's narrated assessment ("it is too late, it's impossible to keep it in") and Perception (Smell)'s sensory catalogue are intradiegetic textual content delivered through the panel. Hawreliak's modal consonance (156–184) names this convergence: multiple semiotic modes addressing the same experiential claim, each independently capable of communicating bodily incapacity, all arriving simultaneously. The convergence reifies the procedural outcome. A failed dice roll becomes a moment of characterisation about the detective's physical ruin, and the experiential intensity exceeds what any of the four channels achieves alone (Hawreliak 124–132 on procedurality requiring reification through other modes).

The morale and health bars are discernible in Salen and Zimmerman's sense, since their state is visible on screen, but the cost a given choice carries is not made perceptible at the moment that choice is taken (34). The bars sit in the corner and report condition continuously, and the interface does not prompt their reading at the points where their state matters most, such as a dialogue option about to deplete morale or an encounter about to cost health. The cost itself remains integrated, accruing toward foreclosure even when the display goes unread. This disjunction, between a consequence that is fully integrated and a state that is visible yet unattended at the point of decision, is the structural condition under which P03's morale death becomes possible.

Likert scores for UI feedback (item 6) showed a wide spread: P02's high score (7) corresponds to the strongest engagement with check-resolution events, their micro-narrative observing that "the UI did make it clear that I managed to get something right." P01's lower score (4) reflects a different reading: their micro-narrative emphasises textual and vocal feedback over visual indicators, locating the panel's content as the primary communicative surface and treating the broader UI as secondary. P03's morale death is the strongest evidence for the discernibility/integration disjunction. Their micro-narrative describes engaging dialogue choices out of narrative curiosity ("I wanted to experience the game fully and I felt skipping the options would make me miss out") with no mention

of monitoring the morale bar. The bar was visible throughout. It did not become legible as a warning at any of the moments preceding the death.

3.3.2 Textual Framing

When it comes to textual framing, three important elements are distinguished. The text panel operates as a polyphonic consciousness, structured around the simultaneous and unsubordinated commentary of the protagonist's twenty-four internalised skill voices in the Bakhtinian sense (Vella and Cielecka 91–102). The voices that speak in any given moment are conditional on character build, so the same conversation is realised as different scriptons (the strings as they appear to the reader) for different players (Aarseth, *Cybertext* 62), with the upstream agency for that conditioning located in the configurative-constructive dimension of Bódi and Thon's typology (157-190). The textual surface operates in modal consonance with other resources at moments of confirmed expectation and in modal irony when the text narrates what the other resources have failed to communicate (Hawreliak 156–184; 220–246). The configuration produces the highest perceived-agency density in the dataset and supplies the structural precondition for the narrative-superposition gap form.

The text panel consolidates every textual channel into a single column and differentiates its speakers through semiotic resources of differing affordance (fig. 3.5). NPC dialogue, the protagonist's internal monologue, environmental description, and the player's own options all reach the player through the one right-hand column. Colour carries the work of parallel identification, marking which voice holds the floor at a glance. The speaker label fixes that attribution explicitly, naming the skill or character. Verbal register supplies the characterological substance, the manner of thought particular to each voice. Typographic weight does the column's organising work, setting NPC names in bold above their lines and bracketing the difficulty notation of a passive check ("CONCEPTUALIZATION [Medium: Success]" in fig. 3.5). The portrait in the upper-left corner shifts to whichever voice currently holds the floor: Cuno's painted face when he speaks, Cunoesse's when she interjects, an icon for whichever skill is commenting. Player options sit at the bottom of the current text block as a numbered list. External speech, internal commentary, and available actions coexist in the same column, and selecting any option requires reading all three.

The structural form of the panel is polyphony in the sense Bakhtin specifies for the novel: independent voices commenting on the same events without being subordinated to a single authorial perspective (xx–xxii). Schirrmacher's intermedial analysis of musical polyphony in literary form clarifies the structural principle at work: the contrapuntal model that polyphonic composition supplies to the novel operates through a dialogic structure of question and response, assertion and contradiction, producing ambiguous oppositions and mirroring inversions across several hierarchically structured levels (Bruhn et al. 1118–1120). Schirrmacher notes that the linear novel cannot reproduce musical simultaneity directly and must approximate it through other means, including the

semantic ambiguity of words and the plurality of contrasting voices. The DE text panel is a further intermedial step within this lineage. The novel can only approximate simultaneity through sequence; the panel comes closer to presenting its voices at once, framed as a single multi-voiced consciousness perceiving one moment. In Elleström's terms, the colour and the name-marker that distinguish a voice are registered sensorially as differences on the screen surface, and their binding to a particular voice is a convention the player acquires across encounters, so that a voice's colour comes to function symbolically, read as that voice before its words are parsed (*Beyond Media Borders* 46–53). The verbal register supplies the evaluative substance the visual resources cannot carry. Visual Calculus narrates with measured clinical precision; Inland Empire with associative, sometimes mystical unease; Volition with weary steadiness; Half Light with paranoid urgency and elevated pitch; Drama with theatrical exaggeration. None of these is a stylistic colouring of a single narrator. Each is an independent voice with its own evaluative perspective on whatever the player has just attempted, and the panel's meaning at any moment exceeds the sum of these resources, the polyphonic consciousness that no single resource carries alone. The player reads these perspectives as the contrapuntal exchange Schirrmacher's analysis names.

Which voices speak is conditional on character build. The system contains every skill's commentary for every interaction the build can encounter, but only those skills above their respective trigger thresholds will speak during a given playthrough. Aarseth's distinction between textons (the strings as they exist in the system) and scriptons (the strings as they appear to the reader) names the structural condition (*Cybertext* 62): the panel's textons are constant across players, and its scriptons are conditional on stats. During the Cuno conversation, P01 received interjections from Electrochemistry, Conceptualization, Rhetoric, Hand/Eye Coordination, Reaction Speed, and Encyclopedia. P02 received Authority, Empathy, Rhetoric, Logic, Volition, Drama, and Encyclopedia. P03 received Electrochemistry, Hand/Eye Coordination, Reaction Speed, Drama, Encyclopedia, Logic, and Volition. The three ensembles overlap on only two or three skills across all participants, so the same Cuno conversation reaches each player as a different polyphonic configuration. The protagonist's psychic interior is rendered through a different set of voices for each player, with build choices (Bódi and Thon’s configurative-constructive agency, exercised before play) deciding which voices are present at all. The same Cuno (the same painted portrait, the same bold-formatted abuse) was rendered through three different polyphonic ensembles: P02's panel narrated power dynamics through voices attuned to social hierarchy and emotional state; P01's panel narrated the same scene through a more diffuse set of physical and analytical observations; P03's panel emphasised reactive and encyclopedic processing alongside, at the critical moment, the psychological collapse Volition narrated as the morale bar emptied.

The upstream agency for this conditioning is configurative-constructive. The character creation screen in *Disco Elysium* therefore acts as a first act of textual co-authorship. P01,

P02, and P03 each invested their stat allocation differently, and the panel rendered different polyphonies as a result. The experience of the Cuno conversation is, in this sense, partly authored hours before the conversation occurs.

The text panel's relation to the other semiotic resources is variable. When P01's Endurance check failed at the hanged man, Endurance's narration of the body's capitulation and Perception (Smell)'s catalogue of what the vomit revealed about the previous night's consumption confirmed what the colour-wash, the forced animation, and the auditory channel had already communicated. The textual narration here completes the modal consonance Hawreliak describes (156–184), converting a failed dice roll into a moment of characterisation P01's micro-narrative explicitly recognises: "it was with failure that it was interesting and surprising, because it created a deeper sense of 'reality', where not everything is perfect." When P03's morale depleted during the Cuno conversation, the same textual surface operated in the opposite mode. Volition's narration of despair and Conceptualization's darkly self-aware commentary ("you're still coming up with sentences. That's a step up from total annihilation, right?") arrived without consonant signals from the other resources: the bars showed condition without conveying urgency, the spatial layout offered no obstruction, the temporal design permitted unhurried engagement. The text channel was narrating death while every other channel narrated continuation. This is what Hawreliak calls modal irony (220–246): productive dissonance between modes. The panel disclosed the situation the system actually obtained at the moment when the other resources had ceased to track it. For player perceived agency, this modal irony is consequential; while the textual channel was about to disclose the protagonist's collapse, every other channel signalled continuation, so the player's perceived authorial agency remained high (dialogue options being weighed, narrative shape being directed) at the moment when mechanical agency had narrowed to a single terminating outcome. The textual surface ratified that narrowing only at the threshold of foreclosure, after the gap between perceived authorship and mechanical possibility had become unrecoverable.

Across these moments, textual framing carries what Worthen names narrative superposition: the experiential complexity necessary to produce a sense of narrative consequence even where the underlying branching converges on the same outcomes (“*Disco Elysium* and Narrative Superposition”). The Cuno conversation rendered through three different polyphonic ensembles is, experientially, three different conversations, even where the NPC dialogue, available options, and procedural outcomes are largely shared. The polyphonic-conditional structure is therefore the structural precondition for the form of the agency gap: perceived agency exceeds actual agency because the textual surface produces interpretive density that the underlying branching does not require.

Likert scores for textual framing (item 7) confirm the analytical primacy: P01 scored 7, P02 scored 6, P03 scored 6, averaging 6.3, the highest resource score in the dataset apart from item 2 (expression). P01's micro-narrative locates the source of the maximum score in the polyphony itself: "the internal aspects of the character's personality actively

commented on the situation," foregrounding skill voices as the principal channel of comprehension. P03's micro-narrative similarly identifies the panel as the system's primary mode of address: "the game was responding to my actions mainly through the Inner Thoughts of my character," with the explicit observation that this resonated with their own experience of internal dialogue. P02's score of 6 reflects engagement with the conversation system as the primary site of play; their reflection that "lack of strength and dexterity are actually very influential. This gives weight to all the possible statistics which is great for replayability" registers the configurative-constructive dimension as felt experience, where the design property becomes legible through play.

3.3.3 Spatial Layout

Spatial layout also distinguishes three elements. Bódi and Thon's spatial-explorative dimension covers the agency a game's design affords through the navigation and traversal of its spaces, divided in two: spatial agency, the movement the avatar can perform within a single space, and explorative agency, which concerns how a game's spaces connect and how far their exploration is incentivised (160–161). The yard is a small, high-density space: a five-second crossing holds the hanged man, the footsteps, the broken fence, the trash container, two NPCs, and Kim. Spatial agency here is modest, since Bódi and Thon note that the game's traversal does not go much beyond a point-and-click adventure (167n17). The live element is explorative: the player decides which of the clustered points to investigate and in what order. Bódi and Thon locate explorative agency in the arrangement of a game's many connected spaces; the yard compresses it into one dense space, where the sequence of investigation becomes a small act of authorship over how the scene assembles. What the environment offers as available is filtered through a nondiegetic machine act in Galloway's sense, the TAB highlight that overlays interactable elements with colour-coded outlines, and that overlay is conditional on character build, so the same yard reaches different players as different scriptons. The configuration extends Jenkins' embedded narrative into a conditional form: the environment holds investigative content for some players and not for others, though the underlying geometry is identical.

The camera sits at an elevated angle, looking down onto a snow-covered yard framed by buildings on the left and right, a fence line at the rear, and the hostel wall behind the player's point of entry (fig. 3.9). The hanged man is suspended from a tree near the centre. The surrounding space contains the footstep marks on the ground, a trash container near the left-side building, a broken section of fence on the perimeter, and the NPCs Cuno and Cunoesse in the upper portion of the yard. Kim accompanies the player. The whole area can be crossed in a few seconds.

The elevated isometric view positions the player as a planner surveying a field of action. McMahan analyses this perspective as historically associated with strategic oversight and CRPG conventions, distinguishing it from first-person framings that produce embodied character navigation (67–83); for the player's authorial agency, the perspective conveys a detached operator's point of view aligned with the detective fiction the game inhabits: a

position of observation and inference, separating the player from the embodied experience of being Harry and orienting them as the detective who reads the scene. Nitsche extends the analysis by treating the camera as a structuring element of the play layer of game space (15–16). The DE camera makes every NPC, object, and point of interest simultaneously visible. Visibility, here, is the condition the perspective offers; what visibility includes is the question the TAB overlay then settles.

Holding TAB activates the colour-coded affordance overlay (fig. 3.9): NPCs and manipulable objects in green, investigative traces such as the footsteps in yellow. Green marks an interaction (a person to speak to, a container to open); yellow marks something examinable as evidence. The overlay is a nondiegetic machine act in Galloway's sense: the address moves from the system to the player, with the protagonist outside the loop. Harry does not see green and yellow outlines around objects; the player does, and the highlights report what the system has registered as available given current state. This distinction operates at the system-to-player layer of communication (Galloway's nondiegetic): the player has informational privilege over the protagonist, knowing which interactions are available given current build state. Mechanical agency is the same for both, since Harry can only interact with what the system permits, but only the player sees the affordance map that makes the permitted interactions legible. The player's authorial agency, the directing of which interactions to pursue, runs on information Harry does not have; the operational agency exercised in the game world remains Harry's. The painted, atmospheric environment becomes, through the overlay, a readable map of possibilities, and the form of communication operates as system-side report on a different channel from diegetic perception.

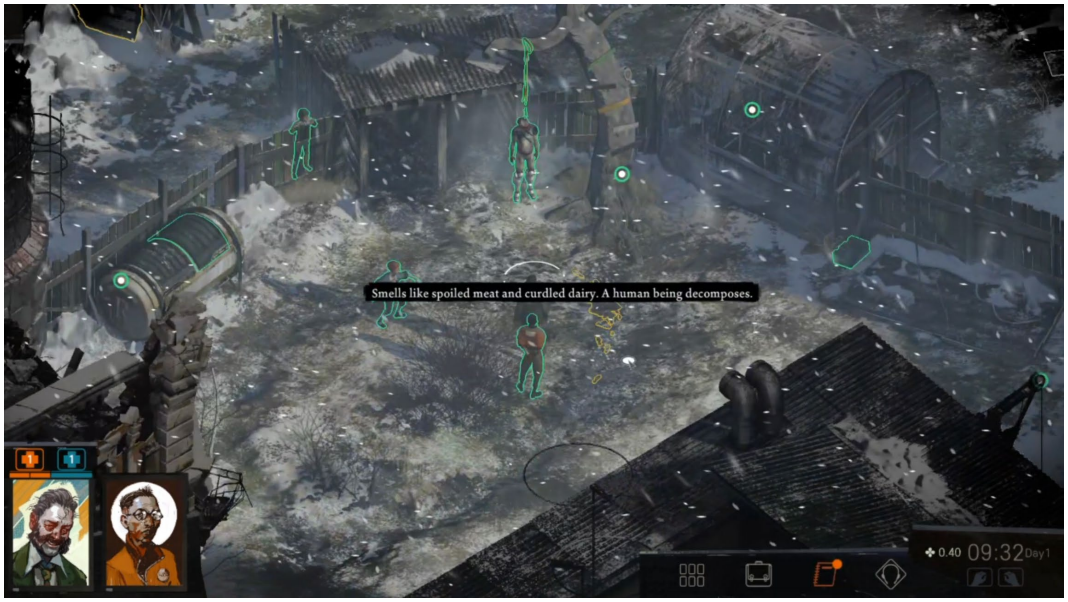


Fig. 3.9 *Disco Elysium*'s Yard layout + TAB-key colour-coded affordance overlay.

What the overlay highlights varies by build. When P01 held TAB, the footsteps appeared in yellow but the broken fence did not register. When P02 held TAB, neither the footsteps

nor the fence appeared; only NPCs and standard objects were highlighted. When P03 held TAB, both the footsteps and the fence were visible. The same key press, applied to the same yard, produced three different affordance maps because the system filters the highlights through the player's Perception stat. The buildings, the tree, the snow, the fences are rendered identically across builds. The interactable layer superimposed on that geometry is conditional. P02 walked through a yard in which the footsteps and the broken fence stayed at the scenery level and never registered as elements of the game.

The yard's textons include all skill-gated interactive content for every possible build; the scriptons available to any individual player are the subset their build permits. Whether a given trace registers depends on whether the protagonist's Perception clears the threshold that converts it from texton to scripton. Below that threshold the trace stays a texton, present in the system but indistinguishable, for that player, from inert scenery. Jenkins' four ways narrative enters games (evocative spaces, enacted narratives, embedded narratives, emergent narratives) all assume that embedded content is uniformly available to anyone who reaches the location. *Disco Elysium's* yard makes the embedding itself conditional. The refinement falls within Jenkins' third category: embedded narrative governed by a player-state threshold that decides whether the content is available at all. Conditionally embedded narrative names this variant, and the texton/scripton distinction supplies its mechanism: the content exists as a texton for every build but becomes a scripton only for builds whose Perception clears the threshold.

Within the highlighted possibilities, spatial-explorative agency operates through interaction order. The yard is small enough that walking offers no navigational challenge, no risk of getting lost, no branching pathways. But within those few seconds, the player has the hanged man (Endurance check), the footsteps (Visual Calculus check, if Perception permits), the broken fence (Perception-gated), the trash container (freely interactable), Cuno and Cunoesse (dialogue with conditional skill-check options), and Kim (ongoing companion dialogue with white skill-checks). The spatial-explorative dimension names the agency; the player's authorship lies in choosing which interactions to undertake and in what sequence, since each interaction may consume time, alter resource state, or close other interactions off. The yard concentrates interactable density, and the order in which interactions are undertaken constitutes the spatial-explorative agency the dimension specifies.

Clicking on an NPC or object repositions the player character to a fixed point and triggers a slow camera pan toward whatever has been clicked. During the Cuno conversation, the camera pans between Cuno (centred) and the player character, while Cunoesse remains at the periphery. Cunoesse’s positioning communicates her status as interference; the camera frames her outside the dyadic exchange between Cuno and the player character. The repositioning is scripted and does not depend on where the player was standing before the click. The play-layer in Nitsche's terms (the layer at which player choices are framed and constrained by the system's spatial responses) is therefore not continuous with

the geometric layer. The player navigates the yard freely, but the camera asserts its own spatial logic at the moment of interaction.

Likert scores for spatial layout (item 8) were P01: 6, P02: 5, P03: 7, with P03's maximum the analytical centre of the closing claim. Their micro-narrative explains the score: "I did not feel physically forced by the game to follow a specific path (for example by putting some limitations of where I can go or who I can talk to before I examined the body)." P03 felt free precisely because the spatial layout did not constrain their movement. Their session ended in morale death during the Cuno conversation, before they reached the body. The freedom the spatial resource communicated was real at its own register: every spatial affordance the system had marked for P03's build was available, and P03 used the affordance freely. What the spatial resource did not communicate, and was not designed to communicate, was the resource-management consequence of using that affordance in a particular order. Spatial freedom and morale depletion belong to different layers of the system, and the disjunction between the layers is a structural condition for the gap form.

3.3.4 Temporal Design

Three elements organise temporal design in this episode: coordination-time as the temporal frame, the white/red distinction within the skill-check system, and the morale and health systems running underneath on a different temporal logic. The clock advances only when the player acts, putting the episode's temporal frame in what Zagal and Mateas call coordination-time, where game-world time is subordinated to player input and does not progress independently (844–868). Within that frame, the white/red distinction in the skill-check system maps onto two different temporal experiences of failure: deferral (white check failed, retriable later) and finality (red check failed, permanently closed), giving the temporal-ergodic dimension two parallel registers within a single resolution system. Two timeframes therefore run at once: locally, coordination-time lets the player set the pace of each interaction, and globally, the depleting resources impose an outer limit on how long the episode can stay open, since the session ends the moment morale or health reaches zero, wherever the player is in the scene. This outer limit is the episode's *temporal envelope*. The configuration produces strong local temporal agency in Mateas and Stern's sense (4–5) within a globally closing envelope that the local freedom does not register.

The in-game clock advances only when the player acts. When the player character stands still, time freezes. NPCs may exhibit idle animations (shifting weight, smoking), but the game clock does not progress. Time advances when the player moves through the environment and when the player selects a dialogue option in the text window. During the hanged man episode, the examination of the crime scene proceeds at a tempo entirely controlled by the player's reading and decision-making speed. There is no countdown, no escalating threat that punishes delay, no NPC routine that the player must intercept within a timing window.

The game alternates between periods of spatial movement (the player walking to a point of interest, during which the clock ticks forward) and periods of textual engagement (the player reading dialogue and selecting options, during which the clock also ticks forward, one increment per selection). Between these active periods, the player can pause indefinitely by simply not acting. A player who reads every line of internal commentary and weighs each dialogue option against the others is not penalised relative to a player who clicks through quickly; the slower player gathers more narrative information from the skill interjections that populate the text window.

In the temporal typology developed by Zagal and Mateas, this is a coordination-time frame in which the progression of game-world time is subordinated to player action; it does not advance independently (844–868). The player controls the pace, lingering on a skill interjection, re-reading a piece of environmental description, or considering how one piece of information connects to another before committing to their next action. Atkinson and colleagues describe such conditions as the basis for aesthetic contemplation in videogames, where extended attention to a visual or narrative object becomes possible only when ludic pressure recedes (523, 527–529). Juul's distinction between play time and fictional time (199–220) applies directly: the hanged man episode's fictional time is almost entirely player-determined, and the clock functions as a record of activity, with no constraining force of its own.

Within this coordination-time frame, the temporal-ergodic dimension operates through the rhythm of engagement: how long the player spends with each interactable element, when they choose to commit to a check, and what they do with the time the system makes available. Speed and timing are absent from the dimension because the system imposes no temporal pressure. The yard's interactable density (analysed in 3.3.3) is given temporal extension by coordination-time. A player can read every internal commentary line at the broken fence, then walk to the footsteps and read every Visual Calculus interjection, then approach the hanged man and rehearse the dialogue options before selecting one. Each of these acts consumes one increment of game-world time, but the investment of real time is at the player's discretion. Temporal-ergodic agency is exercised in the modulation of attention; there is no time pressure to negotiate.

The white/red distinction in the skill-check system inscribes two different temporal experiences of failure into a single resolution mechanism. A white check, if failed, can be retried later after the player has acquired new information, items, or skill points; the failure defers resolution to a future moment in the play session, with no temporal window closing. P01 and P02 both reached the body and failed the central Endurance white check; for both, the failure redirected the investigation and required a return to the body later, after obtaining ammonia. The journal system records these deferred checks, maintaining a visible log of unresolved temporal commitments. Red checks are permanent closures. P02's unused Physical Instrument red check for Cuno, had it been attempted and failed, would have permanently closed the option to physically intimidate Cuno. Aarseth's analysis of aporia and epiphany as fundamental dynamics of ergodic art names what the

two experiences are doing: the deferred white check is a small epiphany of "I must return here with X," which opens the investigative trajectory and keeps it active; the permanent red-check closure is an aporia in Aarseth's sense, a moment where a path becomes structurally inaccessible from this point forward in the session ("Aporia and Epiphany" 31–41).

The episode's morale and health systems operate beneath the coordination-time surface and on a different logic. They deplete continuously through dialogue choices, environmental encounters, and time itself, without any temporal signal that links current depletion to imminent closure. Likert scores for temporal design (item 9) showed the widest variance of any item in the DE dataset: P01 scored 5, P02 scored 5, P03 scored 2. The variance maps onto the local/global tension Mateas and Stern's typology specifies (4–5). For P01 and P02, local temporal agency was strong: each interaction was self-paced, each failure was deferrable, and the global temporal envelope (the resource state running underneath) never closed before the local episode ended. Their moderate scores reflect a configuration in which pacing influenced decisions to some degree without becoming a dominant factor. For P03, the global temporal envelope closed mid-episode. Their session was terminated by the morale system during the Cuno conversation, and their temporal experience was one of unexpected termination; the contemplative control the other participants reported was unavailable. P03's score of 2 records the result of a temporal configuration in which the local agency the coordination-time frame appeared to grant was constrained by a global resource-temporal logic the interface had not communicated as urgent.

P03's case is the structural inverse of the white-check epiphany. Where the white check signals "this can be reopened later," the morale death realises an aporia in the strongest sense: a moment where the play trajectory itself reaches an impasse the system does not allow the player to navigate around. The extension of Aarseth's term beyond the interpretive register he originally develops is necessary here as the morale death is a structural aporia in the precise sense that matters for ergodic art: no further configurative move is available to the player from this position, and the only response the system permits is to reload an earlier save. The aporia and the epiphany are not symmetric in DE's design; the system invests heavily in signalling the white-check epiphany (the journal log, the red-strike-through on failed checks, the visible "you may return later" affordance) and invests almost nothing in signalling the resource-depletion aporia. In Mateas and Stern's terms, local temporal agency is well-supported by integrated semiotic communication; global temporal agency is structurally available but communicatively unsupported.

3.4 Intermedial Techniques

Three intermedial registers organise this episode. The text panel operates within the basic media type Elleström names visual text: inorganic, flat, static, mainly symbolic signs on a surface, organised through prose conventions (*Beyond Media Borders* 56–57). The vocal performances of NPCs and skill voices add a register from a different basic media type, auditory text: sound waves heard in a temporal flow (56), borrowing the conventions of voiced characterisation that visual text alone cannot deploy. Cinematic conventions intervene at scripted moments, most visibly in the close-up that overrides the default isometric camera at the failed Endurance check, and a musicological convention intervenes through the strategic withdrawal of non-diegetic score at the backyard. Rajewsky's media-blindness caution applies throughout: *Disco Elysium* does not become a novel, a film, or a play through any of these references. The analytical task is to specify what each borrowing does within the game's own media-specific means ("Potential Potentials" 24–25).

The analysis operates within the as-if structure of intermedial reference Rajewsky theorises (discussed in 2.2.3): *Disco Elysium* cannot literally become a novel, a film, or a play, but it can write, frame, and stage as if it had the instruments of those media at its disposal. Text in games is unremarkable in itself, since most games deploy text for dialogue, exposition, and instruction; what distinguishes *Disco Elysium*'s text is the deployment of characteristically literary devices (extended interior monologue, polyphonic distribution of consciousness across stylised narrative voices, free-indirect access to the protagonist's perception, descriptive prose density and length) that differentiate it from the dialogue-and-exposition text common to the medium. The skill voices populating the text panel are the most extensive instance of this evocation in the episode, and the close-up at the vomiting moment is the most visually compressed one.

The right-hand panel adopts the conventions of the printed page: prose paragraphs, speaker attributions, top-to-bottom reading alignment, and distinct typefaces for different functions. The player reads prose that includes environmental description, interior monologue, dialogue, and skill commentary, all within a single column. The polyphonic structure 3.3.2 analyses operates here as a layered intermedial inheritance. The DE panel evokes the contrapuntal architecture of the novel, and the novel's contrapuntal architecture is itself an intermedial reference to musical polyphony. The panel therefore makes a double inheritance perceptible at one reading surface, evoking the literary form that had already inherited the musical one. Morozova captures a parallel property from a generic angle, describing the game as a synthetic genre in which interactive literature, CRPGs, and virtual book-games interact (86–97). The intermedial reference here is structural in two registers at once. The panel's reading experience follows conventions familiar from the novel, but it deploys colour-marked simultaneity, build-conditional voice ensembles, and real-time interjection that the linear page can only sequence. The DE text panel is therefore best understood as a further intermedial step in the contrapuntal lineage, evoking the novel through means the novel itself could not deploy.

Every NPC and every skill voice is performed vocally; the player character has none, and this textual silence reinforces the player's position as a reader. The protagonist exists as text the player selects and reads, positioning the player at a hybrid reader-author position, distinct from the performer's position the actor occupies in stage or screen drama (Calleja; Newman). Cuno's aggressive delivery, Kim's measured tone, Inland Empire's unsettled whisper, and Volition's weary steadiness carry characterisation through timbre, rhythm, and affect in registers text alone cannot reach. The vocal register itself draws on multiple traditions: cinematic and animated character voice, the audio-narrative tradition of voiced internal monologue, and (less centrally) theatrical voice performance. The literary register (carried by the silent protagonist's narrated interiority and the prose around it) and the vocal register (carried by NPCs and skill voices) coexist within the same surface. The asymmetry combines two narrative conventions. The panel renders the protagonist in the second person: its descriptive prose and skill voices address a "you" who is named and perceived but never given a narrating voice of his own, the mode of second-person narration (Fludernik 445–479). His interiority reaches the player through internal focalization, the narration restricted to what he perceives and feels without his being its speaker (Genette 189). The skill voices then externalise that interiority as the polyphonic chorus 3.3.2 analyses, so the faculties a first-person novel would gather under a single narrating "I" are here distributed across twenty-four named speakers. The surface combines second-person address with an interiority that is internally focalised yet externalised as voices, a configuration no single literary form holds together.

When the failed Endurance check at the hanged man triggers the vomiting animation, the camera overrides its default isometric perspective to push in to a closer framing of the protagonist. The close-up belongs to the cinematic vocabulary: a film technique for isolating a character's bodily or emotional state and directing viewer attention to physiological detail (Bordwell, Thompson, and Smith 190–192; Balázs 55–56). Although *Disco Elysium*'s default framing is the painterly isometric tableau characteristic of the CRPG tradition, the push-in on Harry's vomiting body imports this filmic convention into the game's visual grammar. The accompanying green vignette draws on graphic-narrative conventions, where colour overlays at the frame edge function as visual signifiers of bodily or emotional state, a convention developed in comics and graphic novels and inherited by games. The visceral sound design draws on cinematic conventions for somatic representation, where close sound design is a standard tool for communicating bodily experience (King and Krzywinska; Ronfard 713–730). The override is what Klevjer analyses as the production of a parallel cinematic space within a game otherwise operating under different spatial conventions (301–309): a scripted moment momentarily reorganises the modal configuration of the play surface so that it operates as a cinematic surface, then returns to the isometric default. Hammeleff Jørgensen's revised four-modality system, derived from Elleström and applied to games, names what the reconfiguration adjusts ("Media and Games" 1–10): the sensorial modality intensifies (visual close range, audible viscera), the structural modality shifts from oversight-survey to narrative attention, and the semiotic modality moves from indexical-affordance reading

toward iconic and indexical signs of bodily state. The "as if" of Rajewsky's account becomes visible in compressed form here as *Disco Elysium* cannot literally produce a cinematic close-up at the camera-distance and angle the genre requires, but the scripted override evokes one through the means available to the engine, and the modal reorganisation is brief enough that the intermedial reference is felt as an intervention.

When the player initiates dialogue with Cuno, the player character is repositioned to a fixed point and the camera pans slowly toward the speaking character, with Cunoesse held at the screen's periphery. Fixed character positions, panning between speakers, compositional use of proximity and periphery to signal relational hierarchy: the staging borrows from the conventions of stage drama and from the cinematic adaptation of theatrical blocking (Wei et al.; Fernández-Vara). The repositioning to a fixed mark before the dialogue begins echoes the stage convention of actors taking marks before a scene. This theatrical reference operates at a smaller modal scale than the close-up's full cinematic override. The geometry of the yard and the isometric camera are preserved; only the conversational arrangement within that geometry assumes theatrical organisation. Rajewsky's "as if" applies in a softer register here: the scene is composed as if it were a stage exchange, with the game's spatial and visual grammar preserved underneath.

The soundtrack composed by Sea Power accompanies the player's exploration of Martinaise with a horn-led ambient piece ("Instrument of Surrender") that establishes a melancholic, atmospheric register for the outdoor environment. When the player enters the backyard behind the Whirling-in-Rags, this non-diegetic music fades and is replaced by diegetic sound only: footsteps on snow, wind, ambient environmental noise. In film, the deliberate removal of underscore at a dramatically significant moment signals a shift in register, typically toward unmediated realism, vulnerability, or confrontation with something the soundtrack cannot aestheticise (Gorbman). The musicological intermedial reference operates through subtraction. *Disco Elysium* withholds its score in the way cinema's scoring conventions have made legible as significance, borrowing the technique of marking gravity by removing the very mode that would otherwise deploy it.

The intermedial references in this episode therefore configure the player's perceived authorial agency through three structural moves. The literary-dramatic hybrid of the text panel produces the polyphonic interpretive surface 3.3.2 analyses, evoking the novel and the dramatic monologue through the panel's own resources. The cinematic close-up override and the silence at the backyard each enact Klevjer's parallel space within a game whose default visual and auditory grammar belongs to a different register, opening brief moments where another medium's conventions reorganise the game's modal configuration before it returns to type. The theatrical blocking of the Cuno conversation operates at a softer register, organising the conversational geometry as if it were stage performance with the game's spatial conventions preserved. The configuration deploys intermedial reference at three intensities: structural for the literary-dramatic surface, compressed and overriding for the cinematic interventions, organisational for the theatrical staging.

3.5 The Agency Gap

P01's micro-narrative articulates the central pattern without analytical vocabulary: "the variability of reactions and text creates the illusion of much more freedom than there actually is." The agency gap in this episode is a network of relations among the four agency dimensions 3.3 distinguishes. It does not reduce to a single distance between actual and perceived agency. The four semiotic resources address each dimension differently, producing tensions where the dimensions diverge and alignments where they converge.

The first relation is the inflation of perceived authorial agency above the mechanical one. Textual framing carries this inflation: the polyphonic ensemble of skill voices (3.3.2) produces experiential density that exceeds the underlying branching, with the build determining which voices populate the panel and the same conversation reaching each player as a different polyphonic configuration. The other three resources align with the textual surface. UI feedback (3.3.1) signals at procedural moments that the player's choices are being processed by a system responsive to the build; spatial layout (3.3.3) delivers the high-density interactable concentration on which authorial selection operates; temporal design (3.3.4) grants the coordination-time within which careful investigative work can be performed. The four resources align in this register to produce what Worthen names narrative superposition, where each of the three participants experienced authorial directing of the narrative even where dialogue paths converged on shared outcomes.

The second relation reverses the polarity, as mechanical agency runs ahead of perceived player agency. P03's morale death during the Cuno conversation, analysed at the level of textual framing in 3.3.2 and at the level of temporal design in 3.3.4, configures a state in which the system was registering cumulative cost the rest of the configuration failed to communicate as urgent. The Health and Morale bars satisfied discernibility, reporting state continuously, but did not prompt their own reading at the moments their state mattered most; the cost they tracked stayed integrated, accumulating toward foreclosure even when the display went unread (Salen and Zimmerman 34). The textual surface ratified the system's reading only at the threshold of foreclosure, when Volition and Conceptualization narrated the breaking point in the parallel cinematic space Klevjer describes (301–309). The four numbered dialogue lines that followed maintained the formal architecture of operational agency while the mechanical configuration had foreclosed any consequential choice; this is the structural aporia in the strongest sense Aarseth gives the term ("Aporia and Epiphany" 31–41).

Across these two relations, perceived player agency, operational agency, and mechanical agency are sometimes in tension and sometimes in alignment with each other and with protagonist agency. During the Cuno conversation, up to the point of the morale collapse, perceived authorial agency aligns with operational agency (the player is selecting dialogue options and directing the narrative) and runs in tension with mechanical agency (the rules are narrowing future outcomes through morale depletion the player does not

read). After the collapse, operational agency aligns with the formal architecture of the four dialogue lines and runs in tension with mechanical agency (the system permits no escape). Protagonist agency runs as a third axis through both phases, since Harry's diegetic capacity to continue conversing is itself the question the morale system is answering at each step, with the answer signalled to the player only at the moment of foreclosure. The agency gap therefore operates as a moving configuration of alignments and tensions across dimensions, with the multimodal orchestration of the semiotic resources determining which dimensions are made legible at which moments.

The third relation is structurally distinct from the first two because the player has no experiential basis for detecting it. P02's Perception stat withheld the footstep marks and the broken fence from the build's possibility space, with the analysis of this conditional embedding given in 3.3.3. The gap here runs between mechanical agency (which the system would permit for a different build) and the perceived possibility space (which is calibrated to what the build can see). Aarseth's texton/scripton distinction (*Cybertext* 62) names what the conditional embedding does to spatial layout: the same physical yard renders as different possibility spaces for different builds, and the system provides no signal that distinguishes content absent from content present yet unsurfaced. Toh's subcategory of ludonarrative (ir)relevance incomplete-information problems names the same condition from the player's side, where the player is less aware that an action is available because the game does not provide the explicit affordance that would surface it (102). P02's awareness of the gap can emerge only retrospectively through comparison with other playthroughs.

Across these three relations, textual framing carries the heaviest agency-gap work in the episode: inflating perceived authorial agency in the first relation, holding the only available signal of mechanical narrowing in the second, and remaining silent on conditional embedding in the third. The continuously visible state indicators remain communicatively inert at the moments their state matters most.

4 *Hitman: Blood Money* Analysis

4.1 Introduction

Hitman: Blood Money (IO Interactive, 2006) is the fourth entry in the *Hitman* stealth-action series, in which the player controls Agent 47, a genetically engineered contract assassin. Each mission gives the player a target and a location, with a sandbox of routes, disguises, weapons, and environmental opportunities to be discovered and combined into a successful elimination. The series' narrativity is largely emergent: although a frame story runs across missions, each mission's micro-narrative is constructed by the player's choices and observed in retrospect, with the post-mission rating ("Silent Assassin" being the highest, awarded for clean and undetected kills) supplying the system's reading of what was accomplished. The mission analysed in this chapter is the first and tutorial-

serving "Death of a Showman," in which Agent 47 infiltrates a derelict amusement park to assassinate its owner, Joseph "Swing King" Clarence.

Tutorial framing is the mission's defining structural feature; a guidance menu replaces the pre-mission briefing that opens every subsequent mission, and objective captions at the top of the screen tell the player what to do next across the entire playthrough. Beneath this instructional surface, the sandbox offers multiple routes, disguises, weapons, and environmental opportunities the guidance does not describe, and the analysis finds that the agency gap takes four distinct forms in the mission, varying across the three participants.

4.2 Mission Overview and Procedural Techniques

The mission "Death of a Showman" is structured by four operational logics: the disguise system, the suspicion and detection system, the NPC positioning and behavioural system, and the Silent Assassin rating system, which evaluates the player's performance at the mission's conclusion. In this context, an operational logic is defined as a computational strategy of representation that addresses both the procedural functioning of a game and the manner in which these operations are communicated to players. The analytical focus of this chapter is on the latter aspect: each of the four logics represents a distinct rule-system that players must infer from observable cues, and the effectiveness of this communication can vary. Bogost's concept of the procedural enthymeme describes this inference process: players observe the visible effects of a rule, infer the underlying rule through interpretive completion, and act on this inferred rule until new evidence necessitates revision. Notably, three participants exposed to the same rule-system may complete the enthymeme differently. Consequently, Section 4.2 will demonstrate that P01, P02, and P03 interpret the same logics with varying operational scope (see Appendix B for a detailed matrix).

Agent 47 begins in his default business suit, which marks him as an intruder to every armed NPC. By incapacitating an NPC and hiding the body, the player can wear the NPC’s outfit, and it allows the player to pass undetected through certain areas, provided they avoid suspicious behaviour in view of NPCs who recognize the disguise as inappropriate. Pape identifies disguise as one of three core modes of stealth, distinct from sneaking and magical stealth, in which the protagonist's visibility is maintained while only their identity is altered (*Aesthetics of Stealth* 173-183). Consequently, the disguise system shifts the challenge from managing visibility to managing recognition: determining which NPCs recognize the disguise as incorrect and from what distance. Among the three participants, the system produced three distinct trajectories. P03 utilized only the disguise suggested by the guidance. P02 discovered the chemist disguise by accident. P01, hindered by guard aggression following early detection, did not reach the disguise stage of play.

NPCs possess cones of vision that extend forward in the direction they face. Actions performed within these cones (running, drawing a weapon, handling a body, or behaving inconsistently with a disguise) raise suspicion; actions outside these cones go unnoticed. When suspicion reaches a certain threshold, the NPC initiates an investigation; if a threat is confirmed, the NPC alerts others, resulting in a cascade of aggression throughout the park. P03 described the system's user interface correlation: "the 'tense' meter would fill up when enemies were on the lookout for me." This meter, located in the bottom-left corner of the screen, tracks the aggregate suspicion level of nearby NPCs. P01 experienced the cascade directly: "when the guards noticed me, the situation quickly spiraled out of control. They would start shooting, and even if I tried to run or hide, the aggression would not subside." Once suspicion escalates to alert and then to combat, no procedural mechanism allows the state to return to un-alerted. The only remaining options are to escape, eliminate all witnesses, or reload a previous save state. The suspicion system functions as a one-way state machine, producing what Aarseth terms aporia: "an absence of possibility" within a cybertext's branching structure (*Cybertext* 3). Although the aporia is local in scope, one alert state can be escaped by reloading, unlike P03's session-ending DE morale death, it is structurally identical: from the alert state, no further configurative move is available within the current playthrough.

Most NPCs in the park either follow scripted patrol routes or remain stationary at designated positions. The primary target, Joseph Clarence, is located in his office on an upper floor of the park's main structure, where he is periodically visited by his secretary, Carol Anne, and a gangster who flirts with her. Three snipers occupy elevated positions: one on the Ferris wheel platform, one on the building platform facing the office, and one patrolling the ground. A sniper rifle, concealed in a case within the drug-lab area, enables the player to eliminate these snipers individually; this timing-gated affordance is explicitly introduced in the guidance menu. P03 identified two specific timing windows: "at the start when you have to distract the guards and by the end, before you're supposed to throw one guard out of the window." Beyond these fixed points, the mission provides additional timing opportunities not described in the guidance, such as when the secretary leaves the office, when the target is alone, or when a disguised NPC can be intercepted without witnesses. The NPC behavioural system operates continuously, regardless of player action, imparting a temporal structure to the game world that is independent of player input and is analysed in detail in section 4.2.4.

The Silent Assassin rating assesses the player's performance at the conclusion of the mission via a post-mission newspaper (fig. 4.1), which presents the results in the form of a journalistic report. This newspaper enumerates killings, shots fired, weapons used, and other statistics, addressing the player from the perspective of the game-world public. The rating functions as the aggregation mechanic introduced in section 2.3 through Sicart (105): each shot fired in "Death of a Showman," each witness left alive, and each unconscious body left in view contributes to a tally that retrospectively determines the rating. In the case of "Death of a Showman," the newspaper displays an anomaly between

its headline and the underlying rating, a relationship further explored in section 4.4. The rating operates on a different timescale than in-mission feedback, retrospectively reframing the completed playthrough as a performance composed of discrete actions whose individual significance was often unclear to the player at the time.

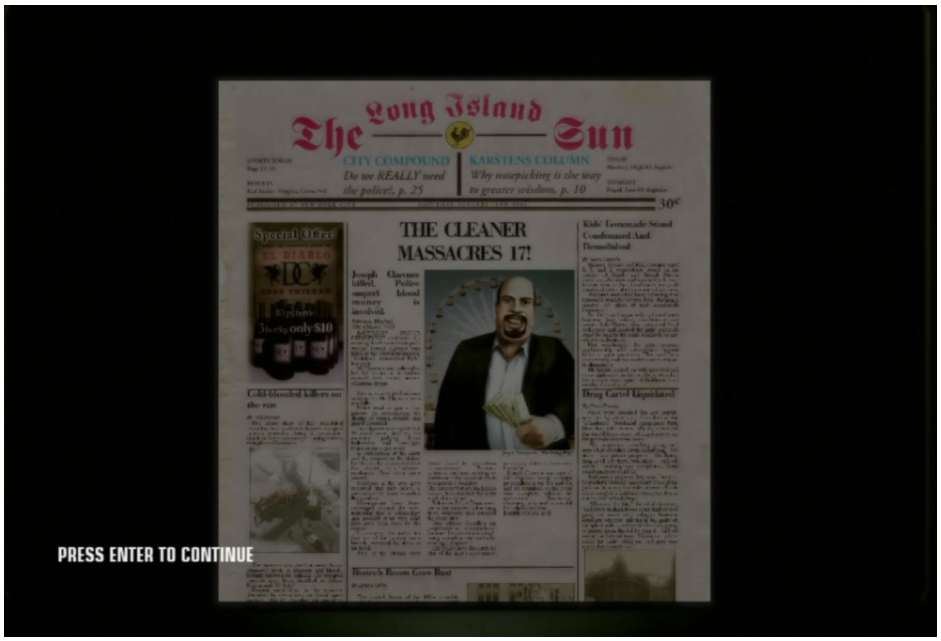


Fig. 4.1. *Hitman: Blood Money*'s post-mission newspaper

The mission affords substantial agency, including disguising, sneaking, distracting, sniping, poisoning, strangling, shooting, pushing from windows, using human shields, sparing witnesses, and escaping. The primary difference among the three participants lies in which of these interactions became perceptible as options during play.

4.3 Semiotic Resource Analysis

4.3.1 UI Feedback

HBM's UI is continuously present and continuously communicative; a HUD whose elements (objective caption, tension meter, radar, control prompts) reports on the system's state without waiting for the player's attention to arrive (fig. 4.2). This continuous-presence configuration narrows expectation management to detection-state communication (Wardrip-Fruin et al., "Agency Reconsidered"). The UI tells the player what the system is registering about NPC suspicion and aggression, frame by frame, but communicates almost nothing about the sandbox's broader procedural depth. Modal consonance (Hawreliak 156–184) concentrates around the detection state: meter, radar, NPC-position colour-coding, and audio cues all converge on the same channel. The system's wider possibility space (alternative routes, alternative disguises, alternative kill methods, alternative timing windows) has no analogous semiotic surface in the HUD that orients the player toward it.

At the top of the screen, a caption line displays the current objective, often phrased as a suggested next action ("distract the guards," "reach the chemist's lab"). Below and to the left, a tension meter indicates the aggregate suspicion state of NPCs in the immediate vicinity, filling from empty (no awareness) through caution (investigation) to combat (active hostility). A compact radar shows NPC positions and facing directions, colour-coded by awareness state. A bottom-right inventory slot displays the currently equipped item. Context-sensitive actions (pick up, throw, use) appear as brief prompts bound to specific keyboard inputs in the top left corner of the HUD.

Punday's account of UI time as a third temporal layer in digital narrative (75–92) applies here in a configuration inverse to DE's; the HUD watches the system continuously and reports the current detection state to the player frame by frame. UI events arrive as continuous reports of system state, generated by the system's own activity, with player-initiated interaction playing a subsidiary role.



Fig. 4.2. *Hitman: Blood Money*'s HUD elements: 1) top left corner occupied by keyboard shortcuts and the interaction associated to them; 2) top centre occupied by briefing/hints; 3) top right corner occupied by navigational radar; 4) bottom left corner occupied by health bar (left) and detection/suspicion meter (right); 5) bottom right corner occupied by loadout selection/in-hand item; 6) the circle in the centre of the screen represents the player's reticle.

A further UI feature distinguishes *Blood Money* from most stealth games of its period: at specific scripted moments, the screen splits along a vertical line, and the right half shows a separate scene of the target (Joseph Clarence) engaged in an activity in his office, while

the left half continues to show Agent 47 in his current position. The first such split-screen triggers as Agent 47 approaches the main gate at mission start, cutting to Clarence in his office for roughly ten seconds before the frame returns to a single view. Subsequent splits trigger at other scripted points during the mission. The feature recurs across *Blood Money* as a whole: MobyGames documents it as a genre-wide convention of the game, describing the device as "split screens [that] show distant actions for timing/assassination hint purposes." The split-screen is a hybrid UI element: diegetically implausible (the player character cannot see through walls), but functionally diegetic in the sense that it communicates what the target is doing in the game world's present tense. In Galloway's typology, it operates as a nondiegetic machine act, delivering information the player needs to plan timing while the protagonist is presumed not to share it.

Item 6 (UI feedback) produced the widest intra-dataset spread of any HBM item: P03 scored 7, P01 scored 5, P02 scored 1. P03's micro-narrative reads the interface as a richly communicative surface ("NPCs in the game would react to the noise I made… the 'tense' meter would fill up when enemies were on the lookout for me") that they used to plan. P02, by contrast, identified a mismatch between the game's control conventions and the conventions of more recent *Hitman* entries they had internalised. In *Blood Money*, for instance, the right mouse button opens the inventory while the weapon-aim function is bound elsewhere. Czerwonka and McArthur identify this as a recognised tutorial problem: when the tutorial's instructional format does not account for the player's controller familiarity and assumed genre conventions, the gap it leaves concerns conventional translation even where the information is plainly displayed (38–40). Additionally, Andersen and colleagues found a tutorial's usefulness to depend on game complexity, adding substantial value in a complex game and little in simpler ones, where players already had what they needed (59–68). The prompts were discernible in Salen and Zimmerman's sense, but discernibility did not by itself make the required action legible (34). P01 sat between these poles, reporting a more specific control-illegibility failure: the UI prompt to use a coin as a distraction was visible, but the mapping from prompt to keyboard input was not, a failure Poretski and Tang describe as the "learnability" problem in videogame UI design (1–4). The same interface produced three different agency outcomes, the difference arising from whether the participant had built a mental model in which its signals were meaningful.

The post-mission newspaper sits outside moment-to-moment UI and delivers summative feedback as a journalistic narrative of the assassination, with photograph, headline, body count, and tally of incidents. P03 singled it out as "an interesting way of giving feedback and statistics to the player." The newspaper is diegetic (it inhabits the fiction of Agent 47's world) and addresses the player with information the protagonist would not encounter about himself, functioning as a nondiegetic machine act delivered through a diegetic form. Its temporal position is also distinct: the HUD reports the system's present-tense detection state; the newspaper reports the playthrough's aggregated history once the playthrough has closed.

4.3.2 Textual Framing

HBM's textual surface is structurally thin, as only the guidance menu, the in-mission objective captions, the context-sensitive inventory prompts, and the post-mission newspaper exist (fig. 4.1). There is no dialogue system (the player character does not speak outside scripted cutscenes), no skill narration, no internal monologue, no environmental description delivered as flavour text. Bódi and Thon split narrative-dramatic agency in two: narrative agency, the capacity to affect predetermined narrative elements and the order in which they unfold (164), and dramatic agency, the narrativity that emerges from ludic events when the player works the mechanics, where those events stay unpredictable and their consequences hold (165). "Death of a Showman" affords little of the first. Its briefing and targets are fixed, and the mission branches no predetermined narrative the player can reorder. The dramatic half is what the sandbox generates, in emergent assassination events that meet Bódi and Thon's criteria, yet the thin textual surface narrates almost none of it while play is underway. The textual surface is busy at moments of instruction and silent during moments of action, and the analytical question this section addresses is what the thin configuration does with the small textual resources it has.

A guidance menu, accessible before the mission begins, introduces the target and the objective. During the mission itself, text appears primarily as top-of-screen objective captions and as context-sensitive prompts at the bottom-right inventory slot. The post-mission newspaper provides a longer stretch of prose, but only after the mission ends.

Item 7 (textual framing) averaged 4.33 for HBM across the three participants (P01: 5, P02: 1, P03: 7), compared to 6.3 for DE. The variance within the HBM dataset mirrors the UI feedback pattern. P03's score of 7 reflects engagement with the guidance captions: "I followed the instructions given on the top of the screen (briefing)." The captions, as a succession of suggested next actions, functioned as a scaffolded narrative of the mission's progression for P03; they read the captions, enacted the captions, and experienced the mission as a legibly unfolding sequence. P02 scored 1 because they experienced the same captions as insufficient: "The tutorial lacked actual information beyond simply telling me to click the button responsible for a given action." The captions told P02 what to do but not how. P01 scored 5, at the middle of the range, and their micro-narrative describes partial engagement: "The game provided clear instructions at the top of the screen… I understood exactly what to do next on a general level." The qualifier "on a general level" marks the divergence from P03. P01 understood the high-level narrative but not the fine-grained procedures, and the textual framing did not bridge the gap.

In their framework, Wildfeuer and Stamenković, assert that tutorial text constructs the discourse structure that organises the player's interpretation of subsequent gameplay; coherence emerges from the alignment of text with what other semiotic resources communicate (28–51). HBM's captions construct a coherent surface discourse (the mission as a sequence of discrete objectives) but rely on the UI feedback, control

prompts, and player's prior genre familiarity to fill in the procedural detail the captions omit. When that fill-in succeeds, as in P03's case, the captions function as a reading surface in their own right and the experience is one of legible scaffolded progression. When the fill-in fails, as in P02's case, the captions reveal themselves as an incomplete translation layer: an instructional surface that names objectives at one level and assumes the player can supply the procedural detail from elsewhere. Toh's ludonarrative (ir)relevance, defines this failure mode as a weak relation between a game's narrative and its gameplay, the case where a player fails to perceive how a message in one module bears on the other, so that the message gains no purchase on their play (99). Toh locates this in the player's reading of the surface, which is why the same caption gave P03 a legible sequence of next actions and left P02 with an instruction whose connection to the required move never registered. P02's case is Toh's "incomplete information" subcategory, where one module does not supply enough for the player to act on the other (102).

The captions stand in modal consonance with other resources at moments of successful instruction: when P03 reads "distract the guards" and the UI prompt indicates the throwable item, when the on-screen colour-coding marks the relevant NPCs, the channels converge on a single experiential claim about what to do next. Where the consonance breaks, the textual surface displays modal irony. The captions claim comprehensive instruction, while the UI prompts assume tacit knowledge of genre conventions and control conventions the captions do not provide. P02's reading registers this irony directly, as the text claims to be teaching the player to play the mission; the player encounters a system that requires knowledge the text does not give them.

The newspaper narrates retrospectively what the player did, in contrast to the captions' prospective register. P03's description of it as "actual data from what I've done in the mission" captures the function. The front page assembles the player's actions into a publicly legible story, addressing the game-world public (the protagonist already knows what happened), standing in for the audience who would encounter the assassination as news. The player, occupying both subject and audience positions, reads their own actions from outside.

HBM's moments of failure (being detected, missing a sniper shot) produce no corresponding text. In DE, the text window could narrate the meaning of a failed Endurance check, with Perception (Smell) describing the vomit and Endurance commenting on the collapse. Here the consequence is delivered through sound (the music shift), through NPC behaviour (turning, shouting, firing), and through the UI (the tension meter filling). Players construct the narrative from non-textual cues during play, or from the retrospective frame the newspaper eventually provides. The narrative-dramatic agency the textual surface could in principle communicate is concentrated at the playthrough's end, in the newspaper's aggregated verdict, away from the moments at which it would shape the playthrough.

4.3.3 Spatial Layout

HBM disperses spatial agency across an amusement park whose physical extent is much larger and whose nested zones of access constitute the mission's possibility space. The spatial-explorative dimension operates here through traversal and access: the player's authorship lies in choosing which routes to take, which disguises to acquire, and which zones to enter, since each zone affords different interactions and risks. The over-the-shoulder camera positions the player inside the park, the disguise system organises the park as nested theatrical zones, and the multiple-route affordance central to Hitman's spatial logic exists procedurally without being signalled by any single semiotic resource. The configuration produces strong perceived spatial agency for players who infer the sandbox's depth and a tutorial-flat experience for players who do not.

The amusement park is rendered in a third-person, over-the-shoulder perspective with free camera control, distinct from DE's fixed isometric view. Agent 47 occupies the centre of the frame, and the player navigates him through the park's buildings, walkways, and open areas using direct keyboard and mouse movement. The camera follows at a fixed relative position, allowing the player to rotate view independently of character facing to scan for NPCs and assess sightlines. The perspective places the player inside the space, supporting embodied navigation that contrasts with the surveying overview DE's isometric camera provided.

Public areas (the main walkways, the food stalls, the queue area) are accessible to Agent 47 in his default suit. Semi-private areas (offices, staff corridors) require disguise. Fully restricted areas (the chemist's lab, the target's inner office) require disguise plus additional stealth manipulation (avoiding actions unnatural to the disguise). The park organises itself into nested zones of access that the player unlocks sequentially, a structure that borrows its spatial logic from theatrical and service-industry dramaturgy. Puente Bienvenido and Sequeiros Bruna apply Goffman's región anterior/región posterior distinction to videogames specifically, treating the on-screen game space as a frontstage where avatars perform and the player's physical space behind the screen as a backstage, with voice channels and household interruptions producing characteristic leakages between the two regions (293–295). Each new disguise unlocks a backstage zone previously inaccessible in the business suit, and each new backstage zone requires the same role-consistent behaviour that the zone before it did. The mission is a lesson in the theatrical dramaturgy of stealth-action design: spatial access is social performance.

P03 scored spatial layout at 6 (Agree), P02 at 5, and P01 at 4 (neutral). P03's micro-narrative catalogues engaged spatial navigation: "I have noticed a couple of environmental opportunities such as using people as human shields, hiding behind objects." P01's neutral score reflects the aggression cascade that ended their stealth attempt: once caught, the spatial layout no longer functioned as a field of possibility but as a channel of pursuit, showing that spatial agency in HBM is conditional on stealth

status. The same environment is either an opportunity surface or a kill corridor depending on the detection state.

On a first playthrough, the multiple-route affordance central to *Hitman's* spatial logic was hard to perceive. P03 reported that "the routes were pretty straightforward in my opinion, I haven't noticed any alternative routes." P02, by contrast, articulated the underlying design without fully executing it: "The tutorial seemed very linear, even though it was clear there were more options to approach it than just those prompted on the screen. I could have used different guns, as well as more stealthy options such as the rope, even in scenes that were clearly laid out for a different approach." P02 perceived the structure but, by their own account, did not execute alternative routes; P03 executed the briefed route and did not perceive alternatives. The same mission contained both readings. What the mission lacked was a semiotic resource that could signal the sandbox's depth to a player who had not already inferred it.

A small room reached by jumping from a balcony houses two security guards playing poker; a scientist eventually enters. With the guidance hints, the player can assassinate the scientist, receiving the disguise outfit and a keycard for further restricted area. P02 arrived at this affordance accidentally, and their micro-narrative describes the moment as the one where the mission's depth became visible: "I managed to change my costume to that of a chemist by pure accident, and it felt like I had just unlocked a whole new range of possibilities." P02 found the chemist disguise without consulting the guidance hints; the spatial layout afforded both routes identically; what differed was which semiotic resource mediated the encounter. P03 reached the disguise by following the guidance captions, and that textual framing recast the scientist's entry as a scheduled beat within a prescribed sequence, a step named in advance. P02 reached the same affordance with the captions bypassed, so the encounter registered through the spatial layout alone, as a surprise the sandbox had concealed. The actual agency was constant, the affordance and the player actions identical; the perceived agency diverged because textual framing mediated the one route and not the other. The same affordance produced incompatible readings of the mission's possibility space.

Jenkins' concept of embedded narrative is relevant in this context and complements the DE extension discussed in section 3.3.3. In DE, embedded narrative is conditionally stat-gated, meaning the environment withholds content from players whose character builds do not allow perception. In contrast, in HBM, embedding is spatially gated; the park distributes narrative affordances, such as disguises, items, and environmental opportunities, throughout its physical layout. The player's interaction with these affordances depends on their movement within the space. Aarseth's distinction between textons and scriptons is applicable: the park's entire set of interactable content constitutes textons, while the player's spatial trajectory determines which elements become scriptons during gameplay. DE filters narrative through perception, whereas HBM filters it through traversal. Together, these approaches define two distinct conditions under which Jenkins'

embedded narrative becomes conditional: gating can occur based on what the avatar perceives or where the avatar travels.

The over-the-shoulder framing positions the player as a third-position observer, between DE's surveying planner of the isometric view and the fully embodied first-person agent of an FPS. Wei et al. characterise the third-person camera's relative flexibility in 3D action and action-adventure games: in its "normal state, the camera follows the avatar at a distance from a slightly higher angle," and "during combat, the camera cranes up and tilts down to enable the player to see more of the environment" (12). The player is close to Agent 47 and not identical with him; they see what he sees approximately, with additional visual latitude from a camera that can rotate independently of his head. The framing supports spatial reasoning through the ability to look around corners while the character remains stationary, and the character's continuous presence in the frame anchors the player's spatial reasoning to his position. This perceptual dynamic aligns with the stealth gameplay loop where observing the world while remaining unobserved is the core activity, a dynamic Pape has theorised as central to the cross-media aesthetic of stealth (*Aesthetics of Stealth*; "Moving in Stealth").

4.3.4 Temporal Design

Time in "Death of a Showman" runs as continuous real-time; the in-game clock advances whether the player acts or not, NPCs walk their patrol routes on their own schedules, and the world's pace is set by the system; the player has no control over it. In Zagal and Mateas's temporal typology, this is a real-world-time frame, the inverse of the coordination-time frame DE's hanged man episode produced (844–868; 3.3.4). The temporal-ergodic dimension operates in HBM through the reading of an autonomous pace. The aggression cascade introduced in 4.1 functions here as a temporal aporia in Aarseth's cybertext sense (3): once triggered, no temporal move returns the mission's state to the pre-detection condition. The configuration produces strong local temporal agency at every moment of decision and a global temporal envelope that closes structurally on detection.

The in-game clock advances continuously, with NPCs walking their patrol routes independently of the player, suspicion accruing whenever Agent 47 lingers in a restricted area, and NPC dialogue (when it occurs) playing out on its own timeline without requiring or allowing player input. Juul's distinction between play time and fictional time (199–220) shows that HBM's fictional time runs continuously, and play time aligns with it under the constraint of physical input speed, where DE's fictional time was almost entirely player-determined. The clock functions here as a constraint upon action, where DE's clock recorded activity without bounding it.

Atkinson and colleagues' analysis of aesthetic contemplation in videogames (523, 527–529) names what real-world-time forecloses. Contemplation requires extended attention, and HBM's continuous flow recedes that condition. The player can pause the game (the

menu freezes the simulation), but the game's design assumes continuous engagement. The contemplative posture DE's coordination-time produced (lingering on a skill interjection, re-reading a piece of environmental description) has no equivalent here: every moment of attention is a moment in which NPCs continue their routines, suspicion meters continue to fill, and timing windows continue to open and close. Temporal-ergodic agency operates through the reading of this autonomous flow, with the player's interpretive labour consisting of identifying when the world's rhythm permits a particular action.

The mission contains several explicit timing windows. Two guards outside the gift shop must be distracted, by a thrown coin, within a brief interval before they return to their vigilant positions. Three snipers must be eliminated one at a time from the SVD (sniper rifle) position before any one of them turns and spots the others falling. Near the end, a gangster can be pushed from an open window onto the street below, an environmental kill that depends on arriving at the right position at the right moment in his routine. P03 flagged two of these: "at the start when you have to distract the guards and by the end, before you're supposed to throw one guard out of the window." Between these explicit windows, the mission also contains implicit ones: moments when the secretary leaves the office, when Clarence is alone, when a disguised NPC can be intercepted without witnesses. The guidance describes the explicit windows, and the implicit ones are available to players who observe the autonomous flow closely enough to identify rhythms the captions do not name.

P03's Likert score for temporal design was 7, the highest in the HBM dataset and higher than any DE temporal score. Their micro-narrative explains the engagement: "the music would change any time enemies were searching for me… the 'tense' meter would fill up." P03 read the temporal flow continuously, using sound and UI cues to gauge whether they had time to act or needed to hide. The high score corresponds to a temporal experience in which the autonomous pace was successfully read, the implicit windows were partially identified, and the local temporal agency available within the real-world-time frame was exercised.

P01's Likert score for temporal design was 4 (neutral), and their micro-narrative describes a temporal failure mode: "When the guards noticed me, the situation quickly spiraled out of control. They would start shooting, and even if I tried to run or hide, the aggression would not subside." The aggression cascade operates as a temporal trap in Aarseth's aporia sense: once triggered, the mission's state cannot be rewound without reloading. The absence of any in-game mechanism to de-escalate an alert state means a single timing failure can end the mission's stealth identity entirely. P01 reloaded to a previous save state to try again. The temporal design communicated, in P01's experience, that mistakes were permanent and that cautious pacing had become a requirement.

Local and global temporal agency operate together here in Mateas and Stern's terms (4–5). Local agency is strong: each second of decision feels consequential, and each window perceptibly opens and closes. The global temporal envelope closes audibly, the music

shift and the tension meter announcing the transition from stealth to alert. The envelope's closure is broadcast at the moment it happens, so the player reads the constraint as it tightens. This signalled closure is the structural condition that makes HBM's temporal design register as constraining, with Item 10 (felt constrained) averaging 6.33 across the three participants.

4.4 Intermedial Techniques

Four intermedial registers organise this mission. Cinematic conventions operate at two scales (continuously through the over-the-shoulder camera, punctually through the split-screen), a musicological register borrowed from film scoring responds to alert-state shifts, a journalistic register frames the post-mission newspaper, and a pedagogical register organises the guidance as scaffolded instruction. Rajewsky's media-blindness caution applies throughout ("Potential Potentials" 24–25): "Death of a Showman" does not become film, journalism, or an instruction manual through any of these references, and the analytical task is to specify what each borrowing does within the game's own media-specific means.

Framing a character from behind and slightly above, positioning the viewer in the narrative space of the character without identifying them with the character's subjectivity: this is the language of action cinema, and the third-person over-the-shoulder camera in Blood Money carries it directly, specifically from the action-thriller and espionage subgenres Hitman draws on. Pape's analysis of the "tracking shot" as a technique for imperceptibility traces the exchange in both directions: stealth videogames adopted cinematic conventions for following a moving body through complex space, and television drama subsequently borrowed the resulting aesthetic back from games (Pape, "Moving in Stealth"). Agent 47, as a character, assembles cinematic assassin archetypes (the precise, minimally expressive professional; the bald silhouette; the tailored suit; the silenced pistols). The camera frames him in ways that make these cinematic resonances legible.

Split-screen, as Szilas and colleagues define it, is "a cinematographic device in which the screen is divided into sub-sections representing different views," and one which games use "only occasionally" because it "may be confusing and disrupts immersion" (Szilas et al. 83). The device originates in cinema, where it establishes visual parallelism between simultaneous actions in different spaces, a technique Simões has analysed as producing a form of narrative polyphony: "several moving pictures… interact with one another on a single frame simultaneously, yet independently, creating… a whole greater than the sum of its isolated lines" (57). Brian De Palma's films and the television series 24 (2001–2010) are frequently cited reference points for split-screen as a suspense device that gives the viewer access to information the protagonist lacks. When "Death of a Showman" splits the screen at the main gate to show Clarence in his office, the game cites this cinematic convention directly; the player sees, as the cinematic viewer sees, a scene inaccessible to the protagonist, timed to produce dramatic simultaneity between approach and target

activity. MobyGames characterises split-screens in the game as serving "timing/assassination hint purposes," identifying the mechanic's ludic function. The convention is repurposed to communicate actionable gameplay information, inheriting the suspense-building form while adapting it for planning, where its cinematic origin served only tension. The split-screen produces what Klevjer analyses as a parallel cinematic space within a game otherwise operating under different framing conventions (301–309): a scripted moment momentarily reorganises the modal configuration so that the play surface operates as cinematic surface, then returns to the over-the-shoulder default. The structural modality moves from continuous third-person framing to dual-frame parallel staging, compressing two locations into a single moment. The cinematic split-screen polyphony is evoked to do the distinct work of stealth-game intelligence.

Composed by Jesper Kyd and recorded with the Budapest Symphony Orchestra and Hungarian Radio Choir (Kyd), the soundtrack uses orchestral and electronic textures that shift with the mission's alert state: ambient layers during undetected stealth, tense strings during suspicion, full percussion and brass during combat. Summers's analysis of alert-state scoring in stealth games applies here: the template he describes, where the music shifts between undetected, suspicious, and combat states, is the same mechanic *Blood Money* deploys (122–123). P03 identified the score's responsiveness: "the music would change any time enemies were searching for me." The scoring borrows from film, where musical state-shifts mark changes in dramatic register, and adapts the technique to real-time gameplay where the shifts are triggered by procedural events the system tracks continuously. The musicological intermedial reference transmutes the player's current procedural state into an emotional register the film convention has already conditioned the player to recognise.

A photograph, a headline, a byline, a body-text article describing the assassination, and smaller sidebar stories, comprise the end-of-mission newspaper's front page. It can be described as the visual furniture of mid-twentieth-century tabloid or mid-market newspapers, a register consistent with the game's retro-noir framing. The intermedial operation here is fictional journalism inside a fictional world. The newspaper as narrative mirror in which the player's actions are reflected back as publicly intelligible events; journalism is evoked through the game's own resources.

Objectives announced, steps scaffolded, competence demonstrated by successful completion: "Death of a Showman" echoes the scaffolded learning of educational media and borrows its tutorial framing from the pedagogical tradition of instructional manuals. The sandbox affordances exceed this scaffolding, and alternative disguises, alternative weapons, alternative routes remain available whether or not the guidance acknowledges them. The mission inherits the form of the tutorial while exceeding the tutorial's communicative scope. The tension between instructional framing and sandbox affordance is one of the mission's defining semiotic conditions, and it is the source of the agency gap.

The intermedial references in this mission configure the player's perceived agency at four intensities: structural for the over-the-shoulder camera, compressed and overriding for the split-screen (Klevjer), transmutative for the score, and scaffolding for the guidance. The newspaper sits outside this gradient, relocating the playthrough's narrative weight to a public-retrospective register the player reads from outside. Each register does distinct work in mediating between the player and the mechanical agency horizon 4.2 specified.

4.5 The Agency Gap

HBM players generally experienced less authorial agency than the sandbox design permitted, and the agency gap took a network of forms parallel to 3.5's analysis of DE but inverted in direction. The mechanical agency horizon (the sandbox of routes, disguises, weapons, and environmental opportunities the system actually affords) sat above the perceived possibility space for every participant. Protagonist agency (what Agent 47 can do as a contract assassin within the fiction) and the player's operational agency (executing moves through the controls) and authorial agency (directing how the contract is fulfilled) were configured by the four semiotic resources in ways that produced four distinct forms of this perceived-below-mechanical relation, plus one local inversion at the mission's close.

The first form is tutorial-concealed sandbox depth: the mission's mechanical agency horizon includes alternative routes and disguises that the guidance does not name, and player traversal determines which of those affordances become scriptons (4.2.3). P02's chemist-disguise moment exemplifies the dynamic. Their micro-narrative describes the discovery as a revelation: "I managed to change my costume to that of a chemist by pure accident, and it felt like I had just unlocked a whole new range of possibilities." Authorial agency expanded the moment the accidental input took P02 off the guided path. P03 experienced the same gap from the opposite side ("I haven't noticed any alternative routes. I have only found 1 disguise provided by the briefing"): the mission's mechanical depth never registered, and their perceived possibility space contracted to what the guidance presented. The structural condition is the texton/scripton distinction extended to traversal-gated embedding (4.2.3). The gap parallels P02's conditional-invisibility gap in DE (3.5) inversely: there, invisible footsteps under DE's stat-gating produced an unseen possibility space; here, following the guidance route under HBM's traversal-gating renders the sandbox's depth invisible.

The second form concerns perceived operational agency falling below the system's mechanical affordance. P01's micro-narrative names a perceived constraint that arises at the level of controls: "Due to the lack of explanations and the severe consequences of mistakes, my freedom of action seemed limited. Instead of experimenting confidently, I acted cautiously and uncertainly." The system afforded freedom (mechanical agency was wide), whilst the interface limited the player's enactment of it (operational agency was narrow). The prompt was visible, but its discernibility did not extend to the mapping from prompt to input that P01 needed to read. A parallel micro-gap operates at the level of

individual mechanics. P03's coin throw was executed successfully as an input but did not produce the expected output, because the specific NPC (a woman in the Swing King's office) was coded not to respond to distractions. P03 had completed the enthymeme "the coin distracts NPCs" from earlier instances, and the system's silent non-response falsified the enthymeme without explaining why. The gap was small in scope, but the configuration is the same: operational agency falls short of mechanical agency at the moment when an inferred rule's scope is not what the player has reasoned it to be.

Across these forms, perceived player agency and mechanical agency are systematically misaligned, while operational agency moves between alignment with the controls and tension with the procedural rules they invoke. During the guidance phase, perceived authorial agency aligns with the captions' instructional discourse, and operational agency aligns with the mapped inputs; mechanical agency, meanwhile, runs in tension with both, since the system permits considerably more than the guidance names. During accidental discovery (P02's chemist moment), perceived authorial agency briefly aligns with mechanical agency, the player recognising affordances the guidance had hidden. Protagonist agency runs as a third axis throughout: Agent 47's diegetic capabilities (silent kills, disguises, environmental hazards) exceed what any one playthrough realises, and the gap between protagonist agency and perceived authorial agency is the structural condition under which HBM's sandbox formally constitutes itself as a sandbox even when a given player experiences it as a corridor.

The fourth form runs in the opposite direction from the others and is local to the mission's close. For every mission after "Death of a Showman" in *Blood Money*, the post-mission newspaper's headline varies with performance; in this mission, the headline reads "Silent Assassin" regardless of the rating the system has computed, because the tutorial's guided path involves killings that would disqualify the player from the Silent Assassin rating in every other mission. The semiotic resource designed to close the loop between action and summative judgment closes it falsely. The configuration is what Hawreliak calls modal irony (220–246): the textual surface communicates one verdict while the procedural surface registers another. The headline-case introduces a perceived-exceeds-mechanical sub-relation within a mission whose dominant pattern runs the other way, showing that gap directions can coexist within a single game and that the four semiotic resources can operate in different directions simultaneously within the same mission.

Across the four forms and the local inversion, the configuration produces a consistent gap signature: perceived authorial agency falls below the mechanical agency horizon for every participant in this dataset, with the four semiotic resources doing the work of this deflation through different combinations of communicative narrowing (the captions, the controls), traversal-gating (the spatial layout), and modal irony (the newspaper). Item 10 (felt constrained by interface, timing, or controls) averaged 6.33 for HBM and 2.0 for DE, and Item 1 (felt free to decide) averaged 2.33 for HBM and 6.0 for DE. The semiotic resources, especially UI feedback and the control-prompt system, carried less of the mechanical agency than the design afforded. The gap concerned what the player

understood they were allowed to do. P02's observation that "the tutorial seemed very linear, even though it was clear there were more options" captures the tension the mission cannot resolve through its own resources: a player who reads the tutorial as authoritative shrinks the mission to the tutorial's size; another player who reads it as advisory expands the mission to the sandbox's size. Both readings are supported by the same semiotic configuration. One structural observation to note is that of the three participants, only P03 reached the target, indicating that tutorial compression in this mission narrows perceived agency for the compliant player and conditions whether the mission can be completed at all for players who depart from or fail at the guidance (see Appendix B).

5 Comparative Discussion

5.1 Introduction

Chapters 3 and 4 analysed the two episodes separately through the same framework, and the two episodes were chosen to represent strongly contrasting design configurations within this framework, ANM, that aims to analyse how any game produces narrativity through the mediation of player agency. This chapter draws the two configurations into systematic dialogue, testing whether the patterns that appeared in one case illuminate the other and whether the analytical vocabulary ANM proposes holds its shape across games with fundamentally different design philosophies.

5.2 Semiotic Resource Configurations

The four-resource framework's value across the two case studies lies in the precision with which it identifies which layer of agency a given semiotic configuration makes perceptible to the player. Agency is not a single quantity that varies in amount across the cases. The mechanical, protagonist, operational, and authorial dimensions ANM distinguishes are activated differently by different resource configurations, and the gap between actual and perceived agency is the site where a given dimension becomes legible to the player as a dimension at all. Comparing DE and HBM resource by resource demonstrates this central claim: multimodal and intermedial configurations do not produce more or less agency, they make different layers of agency available to different kinds of player reading.

Textual framing differs along two structural axes: density (how much interpretive content the surface carries at any moment) and branching (whether that content varies with player state). DE is high on both. The text panel concentrates dialogue, internal monologue, environmental description, and skill commentary into a single column, and the polyphony of skill voices is itself build-conditional, so the panel renders different scriptons for different players. HBM is low on both. The captions, inventory prompts, and post-mission newspaper deliver instructional and summative information at a sparse, non-branching surface that does not vary with what the player has done elsewhere in the mission. The asymmetry foregrounds authorial agency as the dimension at play. DE's text

panel makes authorial agency a perceptual problem: the polyphony of skill voices (3.3.2), the build-conditional rendering of options, and the proliferation of internal commentary force the player to recognise that they are choosing not only an action but a way of being Harry. The gap between what the system actually branches on and what the player feels they are authoring is wide, and Worthen's narrative superposition names how it is felt. HBM's textual surface does the opposite work: the thin, non-branching captions and the retrospective newspaper close off the question of authorship, and the sandbox of alternative routes the system actually affords stays invisible behind the instructional register. Authorial agency is present mechanically and absent perceptually. Likert evidence: DE 6.3 (P01: 7, P02: 6, P03: 6) versus HBM 4.33 (P01: 5, P02: 1, P03: 7), the largest cross-game spread in the dataset. The within-game HBM variance (range 6) tracks the perceptual narrowing: P02 could not read the captions as a complete instruction set, while P03 read the same surface as a scaffolded narrative.

UI feedback differs along the temporal axis Punday's UI-time concept makes legible. In DE (analysed in 3.3.1), the interface is event-bound: the skill-check popup, the colour wash, and the portrait frame intervene at specific moments and then recede. In HBM, the interface is ambient: the tension meter, the radar, the objective caption, and the equipped-item slot communicate the game's state continuously. The asymmetry foregrounds operational agency, the moment-to-moment registration of action as consequence. DE's event-bound UI concentrates operational legibility at procedural climaxes; the player feels each check-resolution as a discrete moment of authored action, while the state of the system between checks remains unsignalled until the next event arrives. HBM's ambient UI distributes operational legibility across every frame; the player feels operational pressure continuously, with the system's wider affordance horizon staying invisible behind the detection-state surface that monopolises the channel. Likert evidence: DE average 5.3 (P01: 4, P02: 7, P03: 5) and HBM average 4.33 (P01: 5, P02: 1, P03: 7); the HBM variance is wider because a continuously active UI offers more for any one reading to miss.

Temporal design differs along two axes that emerge only in comparison: who controls pace (player or system), and whether the global temporal envelope is signalled. DE operates in coordination-time and conceals its global envelope. HBM operates in real-world-time and broadcasts it. The asymmetry foregrounds mechanical agency, what the system can do to or with the player over time. DE's coordination-time hides the mechanical envelope behind player-controlled pacing, and participants who experienced its closure (P03's morale death) read it as a structural surprise after the local agency the clock had granted them. HBM's real-world-time makes the mechanical envelope continuously audible through the adaptive score and the tension meter; participants registered its closure (aggression cascades) as a constraint already named by the system before it occurred. The same mechanical layer, two opposite perceptual surfacings: in DE, the player perceives freedom and meets foreclosure only at the threshold; in HBM, the player perceives constraint at every moment and almost never reaches the foreclosure the

system continuously signals. Likert evidence: HBM 4.67 (a higher average than DE's 4.0), reflecting the temporal channel's continuous communicative work in HBM that DE's coordination-time configuration does not require.

Spatial layout is where the two games converge most closely (DE 6.0, HBM 5.0), with the asymmetry concentrated in gating mechanism. DE filters spatial affordances through perception-stat thresholds; HBM filters them through traversal path. The texton/scripton distinction operates in both, and the resulting invisibility takes the same shape: affordances exist in the system but are not rendered as accessible to a given build (DE) or a given route (HBM). The asymmetry foregrounds protagonist agency, what the avatar can do within the diegetic space. The convergence is the finding here. Both games allow protagonist agency to be conditionally embedded in the environment, and both configurations produce invisible gaps that the player cannot detect from inside a single playthrough. The mechanisms differ; the structure of the gap they produce does not. Likert evidence: DE 6.0 (P01: 6, P02: 5, P03: 7) and HBM 5.0 (P01: 4, P02: 5, P03: 6), the narrowest cross-game spread in the dataset. The intermedial framing also differs: DE's isometric-strategic view aligns with detective-fiction observation, while HBM's over-the-shoulder cinematic framing aligns with action-stealth embodiment.

The four-resource comparison demonstrates that agency in a videogame episode is not a property the system possesses or withholds, and not a quantity the player has more or less of. It is a fluid configuration that emerges, second by second, from a stack of simultaneous mediations: the developer's model of how the world should be readable, the engine's mechanical limits on what can be represented at all, the hardware and bodily setup through which the player perceives, and the semiotic competence the player brings to the surface they are reading. The four semiotic resources ANM tracks are the channels through which this stack becomes legible at the level of a bounded episode, and the four agency dimensions the framework distinguishes are cross-sections of a single fluid process. The actual/perceived agency gap is the visible margin where that fluidity leaves a trace. DE makes the trace dense and continuously perceptible: the polyphonic textual surface, in the contrapuntal-intermedial sense 3.3.2 develops, forces the player to recognise, at each moment, that they are reading from inside a multiply mediated event. HBM makes the trace narrow and almost imperceptible: the tutorial register closes off the recognition of mediation, and the player encounters the mission as a sequence of executions rather than as a layered field of authorship. The triangulation of Likert scores, micro-narratives, recordings, and multimodal annotation registers both configurations as instances of the same underlying phenomenon, what Keylin calls the mediality of own action (Bruhn et al. 1455–1459), realised differently by different multimodal and intermedial configurations and perceived differently by different players within the same configuration.

5.3 Intermedial Techniques

DE's intermedial configuration is literary, theatrical, and musical: the text window as a page, the skill voices as the layered polyphonic ensemble 3.3.2 analyses through Schirrmacher's contrapuntal-intermedial frame, the NPC conversation staging as theatrical blocking with fixed marks and slow camera pans, the strategic withdrawal of the non-diegetic score at the crime scene as a cinematic technique for marking gravity (Gorbman), and the cinematic close-up borrowed for the vomiting animation as a localised override of the game's default visual grammar. The intermedial references in DE work primarily on how the player reads: they frame the protagonist's interiority (literary), the social dynamics of dialogue (theatrical), and the affective weight of moments (cinematic/musicological).

HBM's intermedial configuration is cinematic and print-journalistic, with a theatrical substructure: the third-person over-the-shoulder camera borrows from action and espionage film (Pape, "Moving in Stealth"; Wei et al. 12), the adaptive score borrows from film scoring adapted to procedural triggers (Summers; Kyd), the split-screen cutaways borrow from the cinematic device Szilas and colleagues define as a screen divided into sub-sections of different views (Szilas et al. 83) and Simões reads as a form of narrative polyphony (57), the end-of-mission newspaper borrows from print-journalism iconography, and the frontstage/backstage spatial organisation borrows from theatrical and service-industry dramaturgy (Puente Bienvenido et al. 293–295). The intermedial references in HBM work primarily on how the player acts: they frame Agent 47's embodied movement through hostile space (cinematic), the tonal register of alert states (musicological), the timing of target activity relative to approach (split-screen), the public legibility of the assassination (journalistic), and the social organisation of restricted versus permitted zones (theatrical).

The three shared intermedial resources (theatrical, musical, cinematic) take different analytical shapes in the two games. Theatrically, DE operates at the scene, with fixed marks and a centred camera framing a single conversation, while HBM operates at the stage, organising the park's frontstage and backstage zones into the mission's underlying spatial logic. Musically, DE works by subtraction (withdrawing its score at the backyard to mark gravity) and HBM by substitution (swapping adaptive layers as alert states shift). Cinematically, DE's close-ups and colour-tinted sequences are localised overrides on an otherwise isometric baseline, while HBM's over-the-shoulder framing and split-screen cutaways make cinema the mission's continuous visual grammar. Klevjer's parallel cinematic space (301–309) names the cinematic-intervention moment in both cases (DE's vomiting close-up, HBM's split-screen) as a brief modal reorganisation of the play surface, where DE returns to an isometric baseline and HBM to a cinematic one.

Rajewsky's caution against media blindness ("Potential Potentials" 24–25) holds in both cases, as neither game becomes another medium. Each uses its own media-specific means

to evoke, as if, the practices of. DE's "as if" frames its dialogue branching as the path-choices of a novel's reader; HBM's frames its sandbox as an action film's staged space.

5.4 The Agency Gap Between Both Games

The DE analysis identified three forms of the agency gap: perceived agency exceeding actual agency through narrative superposition, actual agency exceeding perceived agency through unread resource-management consequences, and invisible gaps where the semiotic resources withheld content without signalling its absence. The HBM analysis identified four forms parallel but inverted in direction: actual agency exceeding perceived agency through tutorial-concealed sandbox depth (the sandbox's alternative routes, unseen by P03 under tutorial-following); perceived constraint exceeding actual constraint through aggression cascades and control illegibility (P01's reload, P02's button mismatch); mechanic-level micro-gaps where input and output did not align (the coin that did not distract the woman in the Swing King's office); and the newspaper's fixed "Silent Assassin" headline misrepresenting the rating system's actual verdict.

In DE the gap typically runs in the direction of perceived agency exceeding actual agency, inflating the experience of authorship beyond the system's branching; in HBM, the gap typically runs in the opposite direction, deflating the experience of authorship below the sandbox's affordances. The comparative inversion is the core finding. Both directions are gap states, and both produce divergence between what the system permits and what the player experiences as permitted. In DE, the perceived possibility space exceeds Bódi's actual possibility space; in HBM, the perceived possibility space falls below it. The ANM contribution is to name the direction of the gap as an analytical variable and to show that the same semiotic resources can produce opposite gap polarities depending on how they are configured.

Looking across both datasets, the DE pattern identified as narrative superposition has a structural counterpart in HBM that the second case study allows the thesis to name. Where narrative superposition works by semiotic saturation (textual density makes limited branching feel expansive), HBM produces what this thesis terms *tutorial compression*: the tutorial's instructional register makes sandbox affordances feel like not-yet-permitted content even when they are available. Narrative superposition inflates; tutorial compression deflates. Both are semiotic operations on the player's reading of the possibility space. Both depend on the interaction between specific resources (textual framing in both cases, doing different work) and the player's interpretive habits (reading for narrative richness in DE, reading for instructional authority in HBM). *Tutorial compression* is offered here as a complement to Worthen's superposition, naming the deflationary form the comparative analysis has surfaced.

The clearest shared finding across the two datasets is a pair of invisible gaps: P02's unseen DE footsteps and P03's unseen HBM sandbox alternatives. P02's Perception stat withheld the DE footsteps; P03's guidance-following trajectory withheld the mission's

alternative routes and disguises, which P03 registered as absence ("I haven't noticed any alternative routes. I have only found 1 disguise provided by the briefing"). Aarseth's texton/scripton distinction names the structural condition both cases share: in each, the system's content exists as textons that the player's filtering does not convert to scriptons, with DE's filtering operating through perception (what the build sees) and HBM's through traversal (where the player walks). These gaps are the hardest form for a player to notice during play, because the absence that defines them offers no signal to read; they are typically detected only retrospectively, through comparison with another player's experience or through additional playthroughs.

Within HBM specifically, one gap-form runs opposite to the others. The newspaper's fixed "Silent Assassin" headline retrospectively endorses a performance the rating system has actually rated below the headline's claim, and the gap therefore runs in the same direction as DE's narrative superposition (perceived exceeds actual) within a game whose overall pattern runs the other way. The modal irony here is the textual surface communicates one verdict while the procedural surface registers another. The newspaper-headline case complicates the binary reading of direction: gap directions can coexist within a single game, and the four semiotic resources can operate in different directions simultaneously within the same mission.

Item 1 (freedom) averaged 6.0 for DE and 2.33 for HBM; Item 10 (felt constrained) averaged 2.0 for DE and 6.33 for HBM. The two items run in opposite directions across the games, and the spreads are large enough to support a claim about the games' semiotic signatures: DE communicated freedom via textual saturation even where its branching was narrow, while HBM communicated constraint via tutorial compression and control illegibility even where its sandbox was wide. The gap produced by each game is therefore oriented against the character of the underlying system. Perceived agency is a property of the semiotic configuration through which the design is read.

5.5 Participant Micro-Narratives

The findings drawn together so far span four registers that this section connects through the participant data. The micro-narratives below show how the theoretical, multimodal, and intermedial claims hold up against what each participant articulated, in their own terms, about their two playthroughs. A common finding emerges from reading all three pairs side by side: the participants' interpretive habits were more stable than their agency outcomes, and the games' semiotic configurations met those habits with different degrees of fit.

P01 read for narrative investigation in both games, valuing failure as enrichment in DE ("it was with failure that it was interesting and surprising, because it created a deeper sense of 'reality'") and experiencing failure in HBM as blockage ("the situation quickly spiraled out of control… This effectively blocked further progress"): DE's white-check system defers failure into narrative pacing, while HBM's aggression cascade enforces it

as a session-ending state transition. P01's "illusion of much more freedom than there actually is" articulates Worthen's narrative superposition before the analytical vocabulary names it. P02 read for systemic depth throughout and found it communicated textually in DE ("lack of strength and dexterity are actually very influential… great for replayability") and only through exploratory stumbling in HBM ("by pure accident, and it felt like I had just unlocked a whole new range of possibilities"). For P02, the games differed in whether they surfaced the depth they were looking for; in Aarseth's terms (*Cybertext* 62), the textons were available in both, but the conversion to scriptons depended on different player operations (build allocation in DE, accidental traversal in HBM). P03 read available semiotic surfaces closely in both games, engaging with the Inner Thoughts in DE ("the game was responding to my actions mainly through the Inner Thoughts of my character") and the instructional captions in HBM, yet their DE session ended in morale death while their HBM playthrough completed. Each game's system did different work with the consequential decisions P03 made. In DE, the morale bar was discernible, yet the link between it and the dialogue choices depleting it was never made legible (Salen and Zimmerman 34); in HBM, the captions were discernible, yet they never connected P03 to the sandbox depth they did not name. P03's attentiveness was constant across the two games. The legibility each surface could have delivered was not.

DE micro-narratives emphasise choice, immersion, reaction, and curiosity (P03 on Inner Thoughts as resonance with everyday life; P01 on the variability of reactions; P02 on the influence of statistics). HBM micro-narratives emphasise confusion, constraint, accident, and caution (P02 on control layouts; P01 on lack of explanation; P03 on the coin that did not work). DE's textual saturation produces a vocabulary of agency, whilst HBM's thin instructional surface produces a vocabulary of friction.

6 Conclusion

6.1 Findings

The case-study analyses produced four principal findings. First, the actual/perceived agency gap, previously theorised only in the abstract, can be decomposed through episode-level semiotic annotation into distinct, identifiable forms, each linked to the specific semiotic resource responsible for its emergence. Second, a directional inversion of the agency gap was observed across the two genres. Third, a class of gap was identified that provides the player with no discernible signal. Fourth, agency at the episode level is reconceptualised as a fluid, multi-layered field. Collectively, these findings fulfill the objectives for which ANM was developed and address the gap identified in prior scholarship: establishing a method for connecting the agency gap to the specific semiotic resources that generate it.

The first finding demonstrates that the agency gap can be sorted into a family of specific forms. In the *Disco Elysium* episode, three distinct forms were identified. Perceived

agency exceeded actual agency through narrative superposition, as textual framing created an experiential density the underlying branching did not require, and all three participants identified the polyphonic skill-voice ensemble as the primary source of perceived consequence, even when dialogue paths converged on identical outcomes. Conversely, actual agency exceeded perceived agency in P03's morale death, where the system registered a cumulative cost that the interface failed to communicate as urgent. An invisible gap was present for P02, whose Perception stat kept certain content (such as footstep marks and the broken fence) below the threshold of the experienced possibility space. In all three cases, textual framing contributed most significantly to perceived agency, with a Likert mean of 6.3, the highest among all semiotic resources, while continuously visible state indicators remained communicatively inert at critical moments.

The *Hitman: Blood Money* mission revealed four forms of the agency gap, predominantly in the opposite direction. Tutorial framing obscured the depth of the sandbox, as P02 unexpectedly discovered the chemist disguise and described encountering "a whole new range of possibilities," whereas P03 adhered to the briefing and observed that "the routes were pretty straightforward" with "1 disguise provided by the briefing." For P01, perceived constraint exceeded actual constraint due to the illegibility of control prompts, which limited the freedom afforded by the system ("Due to the lack of explanations and the severe consequences of mistakes, my freedom of action seemed limited"). A mechanic-level micro-gap was evident in P03's coin-throw, where an inferred rule failed when applied to an NPC coded as exempt by the system. The post-mission newspaper further misrepresented outcomes, as its "Silent Assassin" headline appeared regardless of the actual rating, illustrating a modal-ironic divergence between the rating system and its semiotic representation.

The second finding, which constitutes the central comparative result, is the directional inversion of the agency gap between the two games. In *Disco Elysium*, perceived agency generally exceeds actual agency, whereas in *Hitman: Blood Money*, the reverse is true. The same four semiotic resources perform opposite communicative functions in each context. This inversion is reflected in autonomy measures: the "felt free" item averaged 6.0 for *Disco Elysium* and 2.33 for *Hitman: Blood Money*, while "felt constrained" averaged 2.0 for *Disco Elysium* and 6.33 for *Hitman: Blood Money*. Players' interpretations of the possibility space were shaped by the semiotic configuration, as procedural affordances alone would have produced different perceptions in both cases. This inversion also enables a conceptual contribution: *Disco Elysium's* textual saturation generates narrative superposition, while *Hitman: Blood Money*'s tutorial framing produces a deflationary counterpart, termed *tutorial compression* in this study.

The third finding identifies the invisible gap as a distinct and theoretically significant category. This gap was observed in both games: in P02's unrecognized footsteps in *Disco Elysium* and in P03's unnoticed routes in *Hitman: Blood Money*. Aarseth's texton/scripton distinction characterizes their shared structure: system content exists as textons that the player's filtering never transforms into scriptons, with filtering occurring through

perception in *Disco Elysium* and through traversal in *Hitman: Blood Money*. This form is particularly significant because it is the most difficult for players to detect; its defining absence provides no signal and typically becomes apparent only retrospectively, such as through another player's session or a subsequent playthrough.

The fourth finding arises from the comparative analysis in its entirety. Agency within a videogame episode is characterized as a fluid, multi-layered field of interdependencies, transmedial in nature and intermedially configured at the point of manifestation. The operative dependencies at any moment include the developer's model of legible play, the engine's representational constraints, the player's hardware and physical context, their moment-to-moment attention, and the semiotic competence developed through engagement with other media. These factors operate concurrently, shaping both active comprehension and the subconscious organisation of rules and symbols that persist even during lapses in attention. The triangulation of Likert scores, micro-narratives, recordings, and multimodal-intermedial annotation captured this configuration as it varied among players engaging with the same software on identical hardware: P02 and P03 played the same *Hitman: Blood Money* mission, yet, due to differing mediational configurations, effectively experienced two distinct missions. ANM's contribution is to render aspects of this fluidity observable at the level of a bounded episode, with the four semiotic resources serving as channels for observation, the four agency dimensions functioning as cross-sections, and the actual/perceived gap marking the visible margin where the configuration becomes evident.

6.2 Implications and Limitations of the ANM Framework

The findings suggest two refinements to the ANM framework and delineate its current limitations. The first refinement addresses spatial layout; building on Jenkins, the original plan treated it as an organisation of discoverable content within the environment, and both cases forced an extension, as *Disco Elysium* embeds content conditionally via the Perception stat and *Hitman: Blood Money* does so through player trajectory. Drawing on Aarseth's texton/scripton distinction, these conditionally embedded forms were identified as emergent features that strengthen the framework. The second refinement pertains to temporal design, which was informed by a comparatively limited body of game studies literature (Zagal and Mateas; Juul). The analyses produced categories that the framework had previously only outlined schematically, including coordination-time versus real-world-time, deferral versus finality in white and red checks, and aggression cascades as temporal traps. Temporal design thus emerges as the resource most in need of further theoretical development prior to a third case study.

The findings also establish a methodological condition; ANM requires participants to reflect on resources they have encountered but may not recognize as explicit analytical objects, such as tutorial captions, suspicion meters, skill checks, or split-screen cutaways. As a result, the research instruments perform significant translation work, and participant

reports do not always correspond directly to the framework's categories. Researcher observation during the six recorded sessions supplemented these instruments by capturing behaviours not verbalized by participants, including pauses during confusion, repeated interactions with unresponsive affordances, and camera movements indicating disorientation. Canossa and colleagues advocate for this methodological blending, as behavioural indicators of frustration often precede verbal acknowledgment. Translating participant reports into the framework's categories is therefore an essential methodological step, typical of case-study research on technically specialized material and is explicitly acknowledged in this thesis.

The framework's scope is intentionally bounded, and these boundaries inform its future development. The first limitation pertains to the player. ANM conceptualizes the player as an individual interpreter, a simplification that the data already challenge. For example, P02's confusion during the Hitman mission primarily resulted from genre expectations, as the mission's controls differed from those in later Hitman titles previously played by P02. The framework currently treats such effects as noise, whereas they are more accurately understood as manifestations of player history. An expanded ANM would incorporate player disposition, systematically recording prior genre experience and default interpretive habits, thereby allowing these factors to influence which resources a player attends to during gameplay.

The second boundary is particularly significant, as ANM currently operates at the level of a bounded episode, yet both case studies demonstrate that this approach can misrepresent how resources function across an entire game. In *Disco Elysium*, the initial allocation of stats determines what the player can perceive from the outset, meaning that the configuration assumed in an episode-level analysis is itself the result of prior decisions. In *Hitman: Blood Money*, the opening mission is atypical in terms of agency, as it functions as a tutorial that imposes greater constraints and withholds systems that are accessible in later missions. This restricts both the perceived agency and the narrative possibilities available to the player. Extending the framework to encompass full playthroughs, where the four resources accrue significance over time and the agency gap evolves, represents the primary direction for future development. Assessing whether the four resources remain adequate at this broader scale, or require augmentation, is a critical challenge for the framework. Such an extension would also enable ANM to inform contexts beyond literary analysis, including the game development cycle and user research in game design, where understanding how system states become perceivable and actionable over time is a central concern.

The third boundary relates to genre. The four resources were developed based on single-player, authored, non-networked games. Applying the framework to multiplayer games, which introduce social agency, to procedurally generated games, where textual framing and spatial layout are not predetermined, or to virtual-reality games, where user interface feedback and spatial layout merge into a unified perceptual field, would each necessitate genre-specific adaptations. Reflexive and metafictional designs present additional

challenges. For instance, when a game manipulates its own semiotic resources against the player, as in the codec deception in *Metal Gear Solid 2* or the required save deletion in *NieR: Automata*, the act of deception itself becomes a resource. Currently, ANM records what a resource reports, but it would need to account for instances where a resource's intentional failure conveys meaning. The framework's modularity at the resource level and the distinction between actual and perceived agency should remain applicable across these extensions, although each scenario requires further theoretical development rather than immediate application.

The two case studies, each involving three participants, prioritized depth over breadth. The analytical categories remained robust across two markedly different cases, and the distinctions generated by the framework, such as narrative superposition, tutorial compression, invisible gaps, conditionally embedded narrative, and directional inversion of the gap, provide a vocabulary for further analysis. In some applications, the framework requires no additional modification, as ANM fundamentally accounts for how system states become perceivable and can thus specify what is lost when a perceptual channel is removed, a question central to accessibility research. In these respects, the framework is best understood as a humanities-oriented analytical tool for close reading of agency in games. It is explicit about the player dimension it does not yet model and the social and longitudinal aspects it does not address, and is intended to complement, rather than replace, instruments such as Player Experience of Need Satisfaction model (PENS). ANM serves as a net that captures fragments of a dynamic phenomenon, since agency's fluidity exceeds the capacity of any single analytical approach. Elleström's modalities and Keylin's concept of the “mediality of own action” offer additional vocabulary for articulating both what the framework captures and what eludes it. Understanding fluid agency as a property of mediated play is the central contribution of this thesis, and extending this analysis across a broader range of games and players remains an ongoing task.

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Appendix A: *Disco Elysium* Participant Matrix

Procedural event	**P01**	**P02**	**P03**
Central Endurance white check (legendary difficulty; hanged man interaction)	Endurance 2, approximately 3% success, failed	Comparable build, failed	Did not reach the check (died earlier during the Cuno conversation)
Visual Calculus white check at the footstep marks beneath the hanged man	Succeeded; Visual Calculus dominated the subsequent investigation	Did not encounter: Perception below the threshold required for the marks to register as interactable	Visible but failed the Visual Calculus check
Broken fence on the backyard perimeter	Did not interact	Not detectable: Perception below the threshold required for the fence to register	Visible but failed the relevant check
Physical Instrument red check during the Cuno conversation	Not visible (relevant skill below threshold)	Visible (could have physically intimidated Cuno)	Not visible (relevant skill below threshold)
Health system: depletion events	No depletion-related death	Died twice during the prologue under the silent-warning condition (one-point Health, no healing item, no blinking warning)	No depletion-related death
Morale system: depletion events	No depletion-related death	No depletion-related death	Died from Morale depletion during the Cuno conversation; Volition and Conceptualization narrated the breaking point
Episode outcome	Reached the body and ran the Endurance and Visual Calculus sequence	Reached the body on a third attempt after two prologue deaths	Did not reach the body

Appendix B: *Hitman: Blood Money* Participant Matrix

Procedural event	P01	P02	P03
Chemist disguise (a scientist enters the poker room; killed for outfit and keycard)	Did not reach the disguise stage, pinned down by guard aggression after an early detection	Acquired off the guidance: encountered the scientist's body and used the change-disguise input, describing the moment as a revelation of "a whole new range of possibilities"	Acquired by following the briefing in the top-centre HUD: "1 disguise provided by the briefing"
Aggression cascade (one-way state machine; alert-to-combat with no de-escalation affordance)	Triggered the full cascade after an early detection and reloaded; "the situation quickly spiraled out of control… the aggression would not subside"	No cascade triggered	No cascade triggered; read the tension meter as planning information: "the 'tense' meter would fill up when enemies were on the lookout for me"
Control-prompt legibility (coin-distraction prompt and related context-sensitive inputs)	Prompt visible, input mapping illegible: "due to the lack of explanations and the severe consequences of mistakes, my freedom of action seemed limited"	Control conventions mismatched against later Hitman entries (right mouse button binding): "the tutorial lacked actual information beyond simply telling me to click the button responsible for a given action"	Read and followed the prompts: "I followed the instructions given on the top of the screen (briefing)"
Coin-throw mechanic against an NPC coded as exempt	Did not execute the coin-throw (input mapping illegible)	Coin-throw not documented as a divergence point	Coin thrown at the woman in the Swing King's office; the system silently registered no

			response, falsifying the inferred rule without explaining the exemption
Mission outcome	Cascade and reload determined the playthrough's arc; never reached the disguise stage and did not progress to the target	Reached Clarence's office through accidental sandbox discovery but was killed by a guard during the confrontation; absence of a prior save, combined with accumulated control-convention frustration, ended the session without completion of the mission	Reached the target by following the briefing throughout; the coin-throw micro-failure did not interrupt progression

Appendix C: Autonomy Questionnaire (Likert Items, 1–7)

Items (1 = strongly disagree, 7 = strongly agree; items 4 and 10 are phrased to indicate constraint, marked (R) for reverse-coded):

1. I felt free to decide how to approach this scene.
2. I could express myself through different play styles or strategies.
3. My choices had meaningful consequences for the game world.
4. The game pushed me down a single path. (R)
5. I clearly understood the options available to me.
6. The on-screen feedback (icons, meters, highlights, sound) helped me understand the effects of my actions.
7. The words and text on screen (dialogue, descriptions, labels) shaped how I understood my choices.

8. The layout of the environment gave me a sense of where I could go and what I could do.

9. The pacing or timing of events influenced how I made decisions.

10. I felt constrained by the interface, timing, or controls. (R)

Scores by participant and episode (M = three-participant mean):

Item	P01 DE	P02 DE	P03 DE	DE M	P01 HBM	P02 HBM	P03 HBM	HBM M
1	6	5	7	6.00	2	3	2	2.33
2	7	7	7	7.00	2	1	5	2.67
3	6	7	6	6.33	4	2	7	4.33
4 (R)	2	2	2	2.00	6	7	2	5.00
5	5	5	5	5.00	5	1	2	2.67
6	4	7	5	5.33	5	1	7	4.33
7	7	6	6	6.33	5	1	7	4.33
8	6	5	7	6.00	4	5	6	5.00
9	5	5	2	4.00	4	3	7	4.67
10 (R)	2	3	1	2.00	6	7	6	6.33

Appendix D: Micro-Narrative Questions

Disco Elysium:

Reflect on the scene where you examined the hanged man for the first time. Address as many of the following as you can:

• What were you trying to find out or achieve when you reached the body and began examining it?

• What options did you notice during the examination: dialogue choices, skill checks, things to inspect, ways to interact with Kim or the surroundings?

• What did you choose and why? If a skill check appeared, did you attempt it? What motivated your decisions?

• How did the game respond? what feedback did you receive (internal voices commenting, skill-check results, Kim's reactions, changes in the UI or text descriptions)?

• Did anything about the scene surprise you; a moment where the game gave you more freedom than expected, or shut down an option you thought you had?

• How authorial did you feel during this scene? Were you directing the investigation, or following a path the game had laid out for you?

Hitman: Blood Money:

Reflect on the mission you just completed. Address as many of the following as you can:

• What was your main objective during this mission, and how did you decide on your approach?

• What options did you notice; routes through the park, disguises, tools, environmental opportunities, timing windows?

• What did you choose and why? Describe a specific decision and what influenced it.

• How did the game respond; what feedback did you receive (HUD indicators, NPC reactions, alert states, sound cues, camera behaviour)?

• Did anything about the mission surprise you? A moment where the game gave you more freedom than expected, or restricted you in a way you didn't anticipate?

• How authorial did you feel during this mission? Were you crafting your own plan, or being guided through the level step by step?